

Light Nuclei Formation in Heavy-Ion Collisions



**Facility for
Rare Isotope Beams**
at Michigan State University

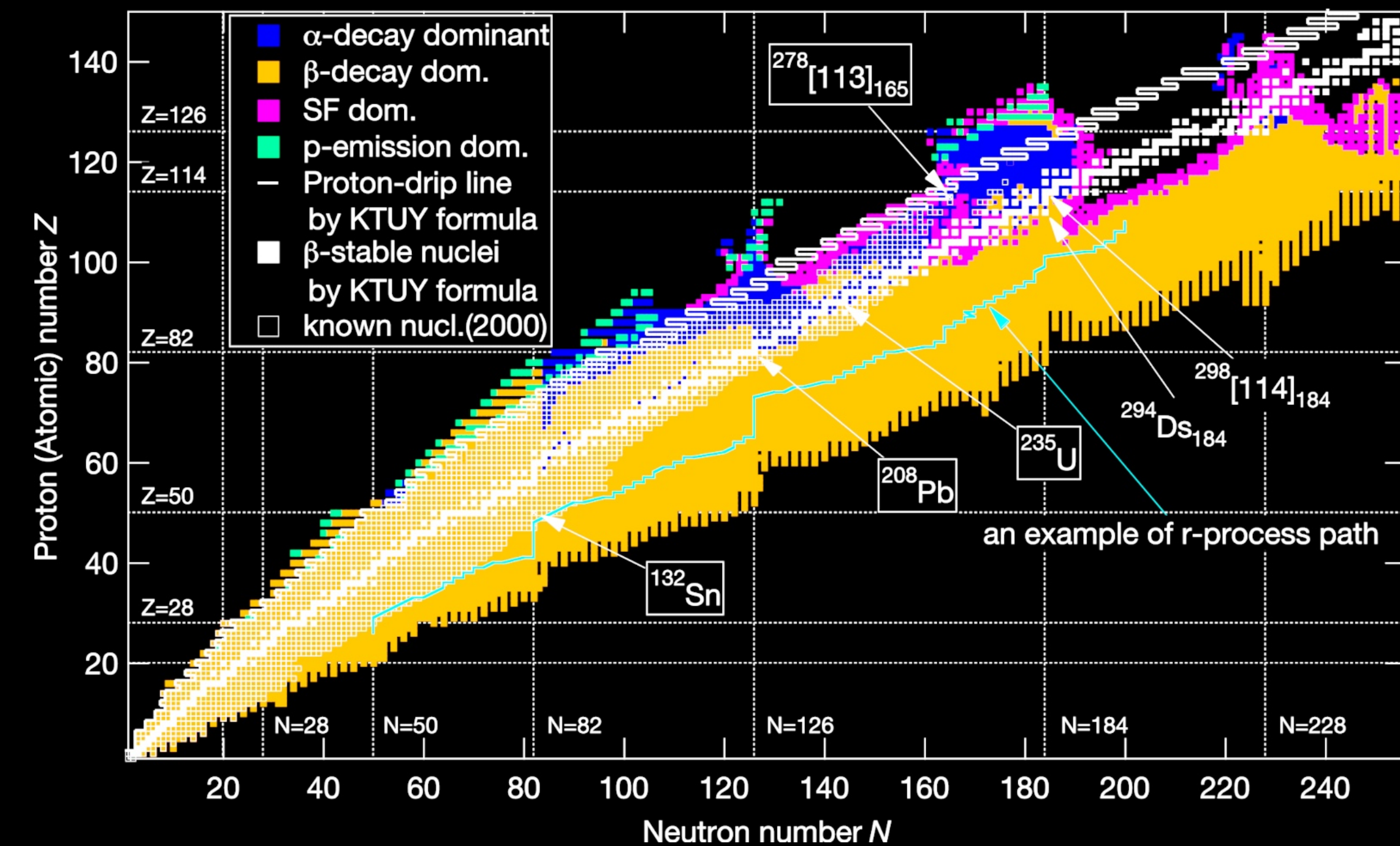
Oleh Savchuk
TMEP meeting, July 15th

Nuclear matter

Interacting protons and neutrons

2

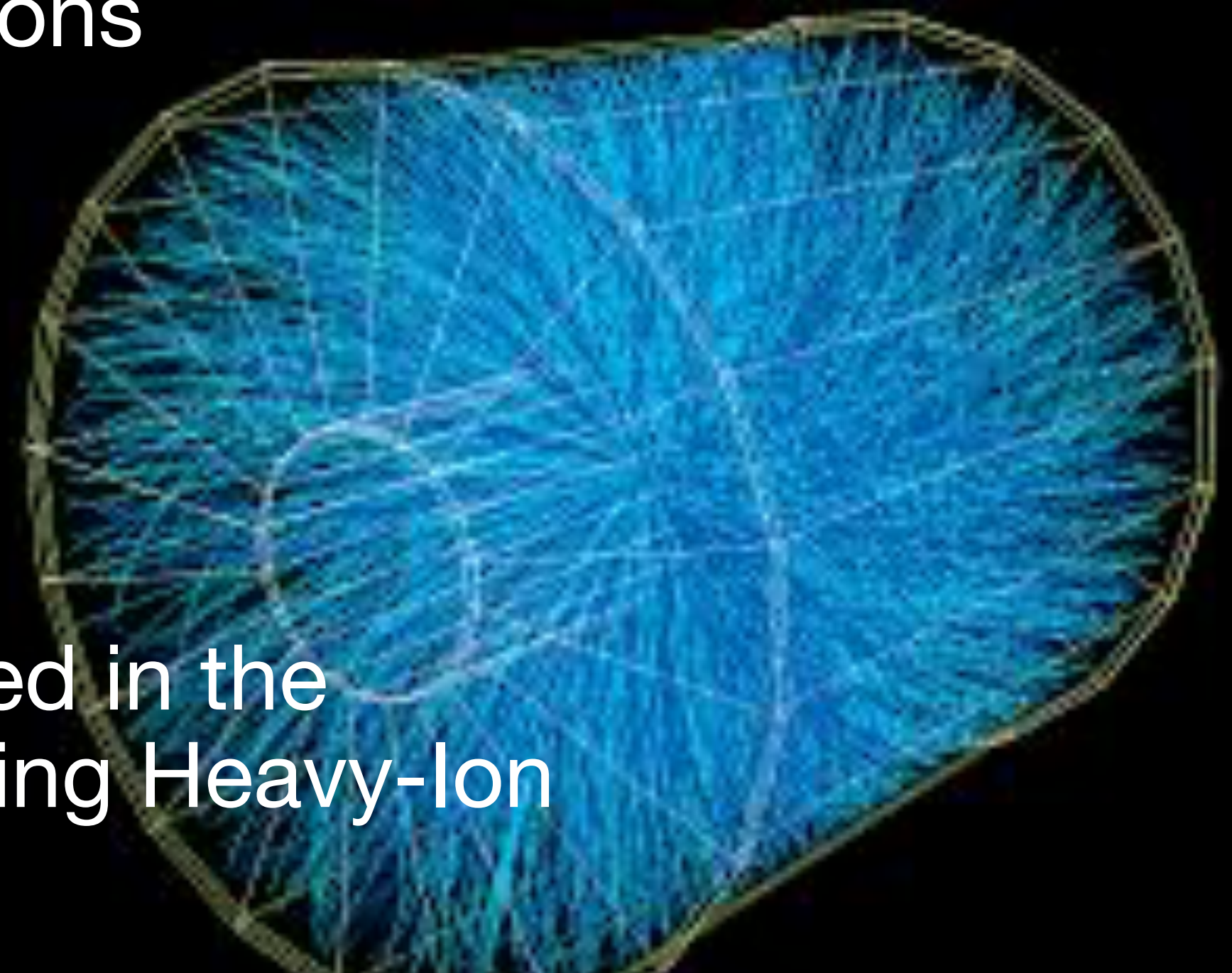
Constrained at “normal” conditions from nuclei:



(Binary) Neutron stars and mergers

Poorly understood under “extreme” conditions

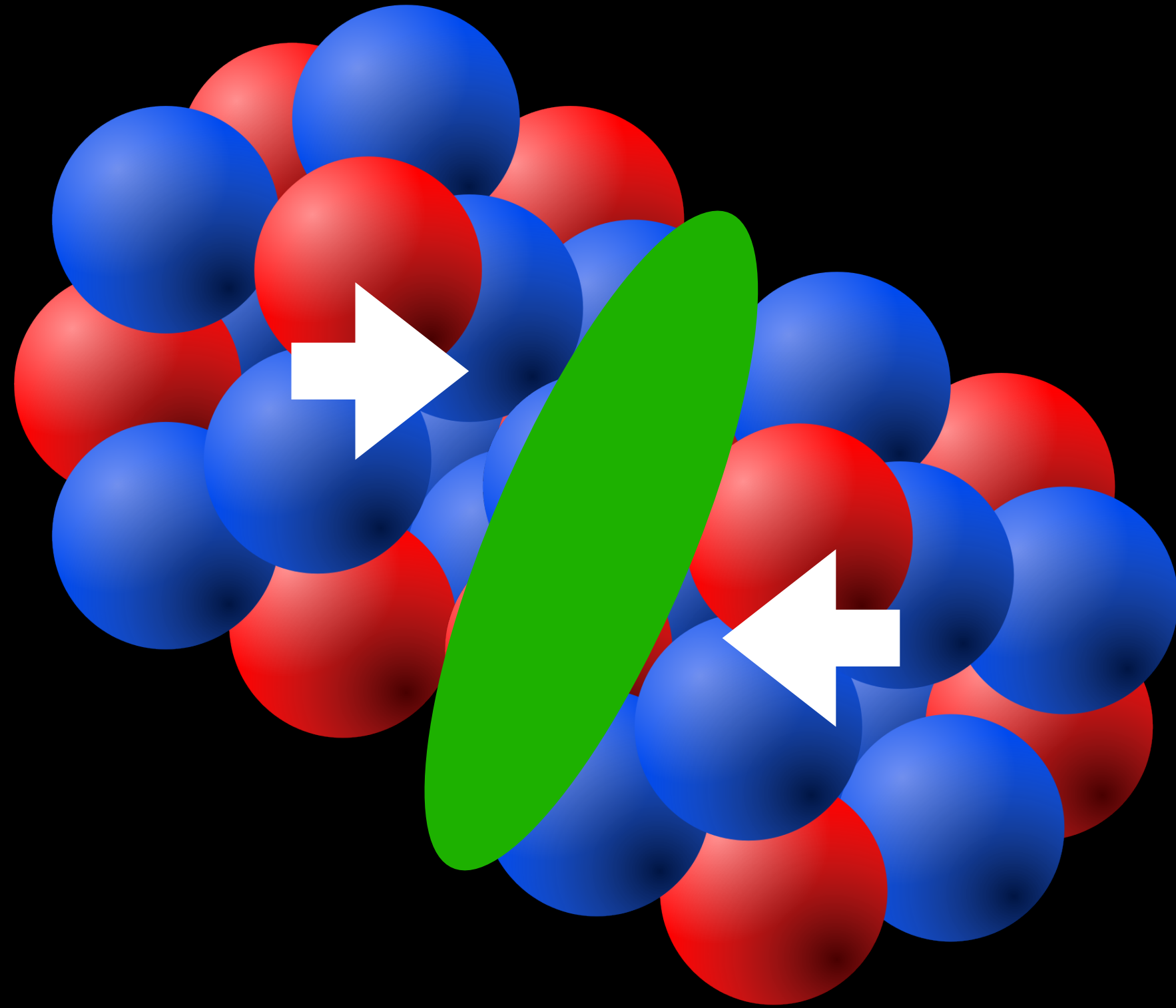
Can be studied in the laboratory using Heavy-Ion Collisions



Heavy-Ion Collision

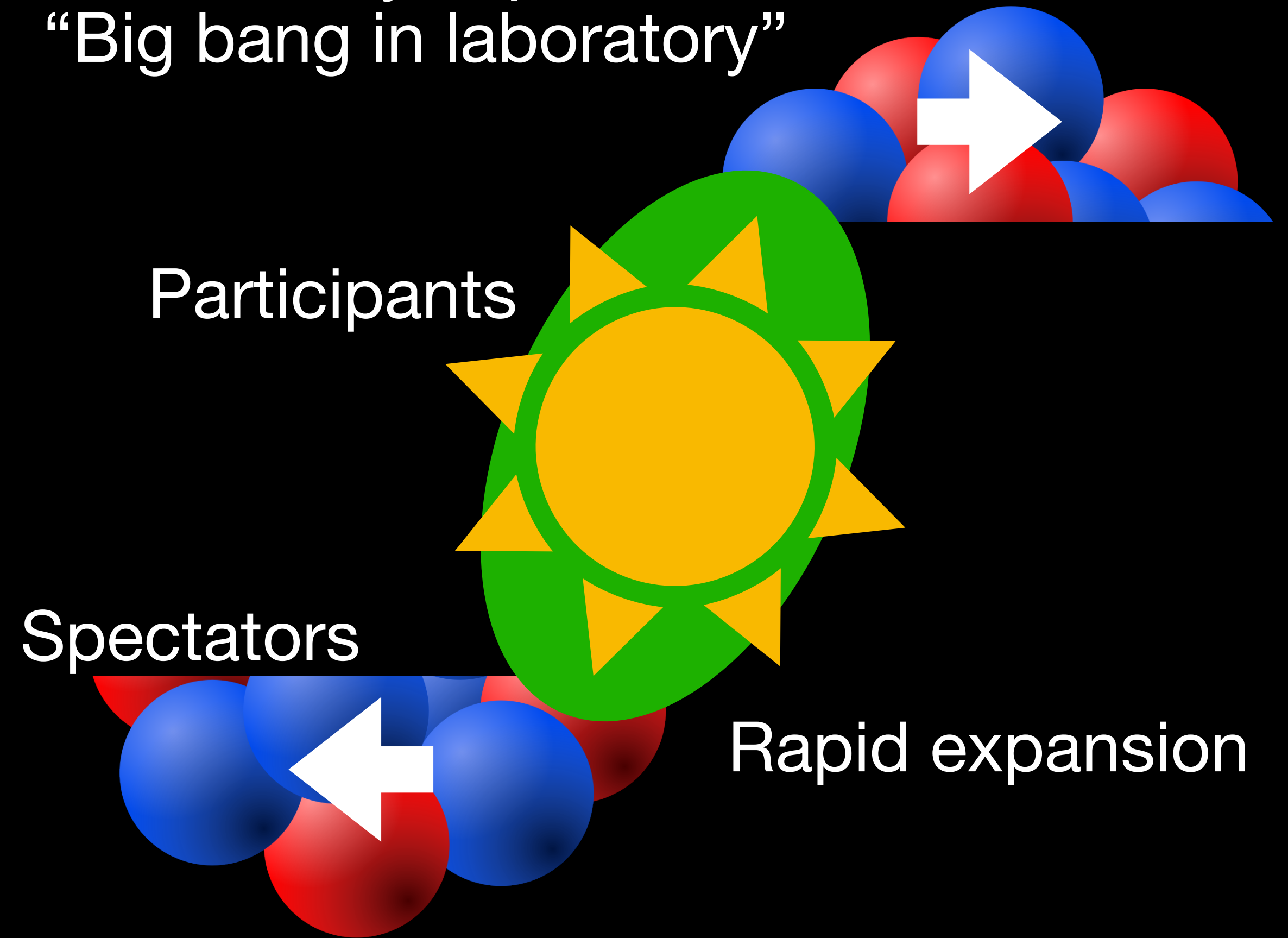
3

Deposition of energy into overlap region



Large energy and particle densities

Creates tiny explosion
“Big bang in laboratory”

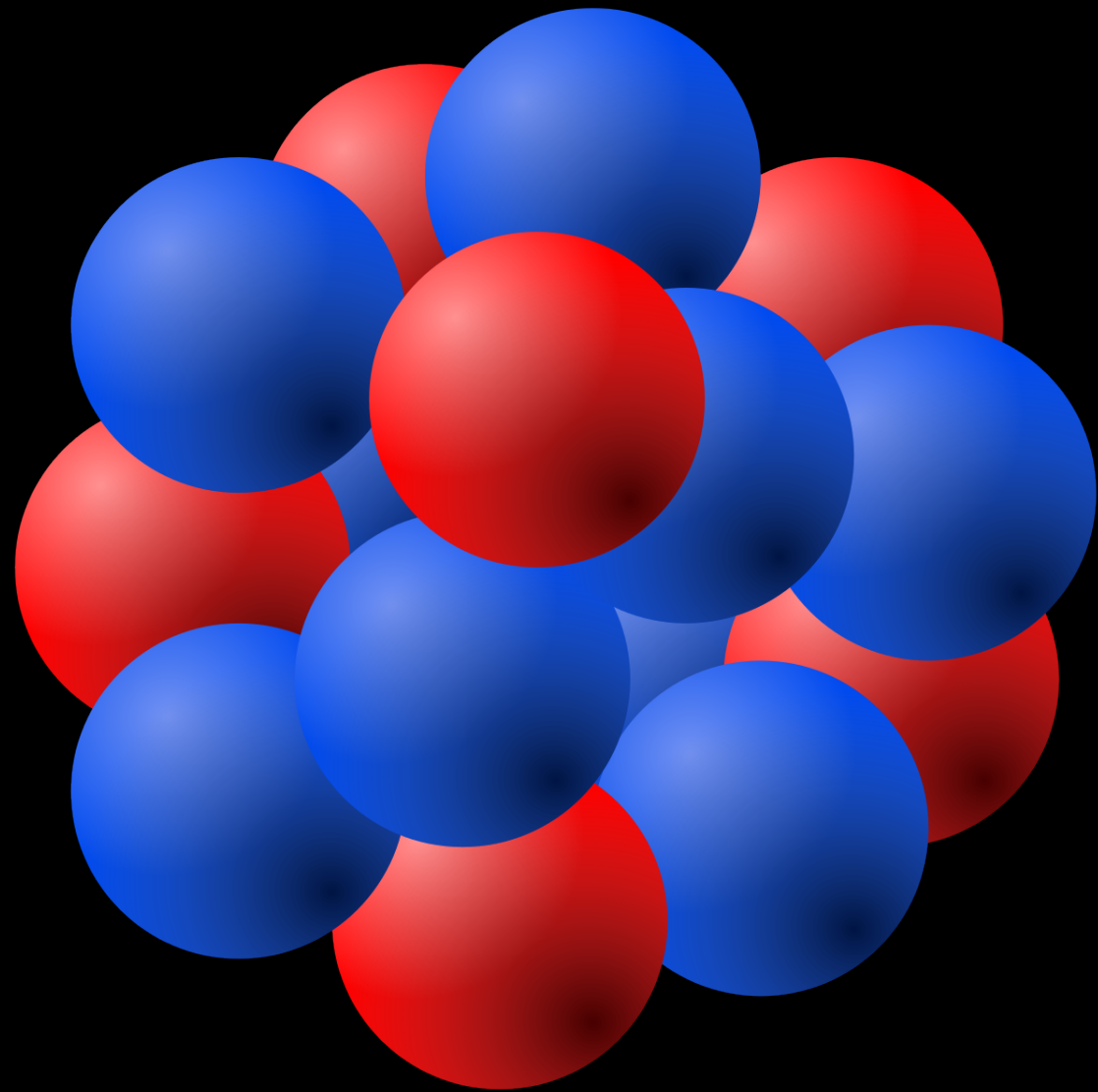


Collective motion and local “temperature”

Stages of Heavy-Ion Collision

4

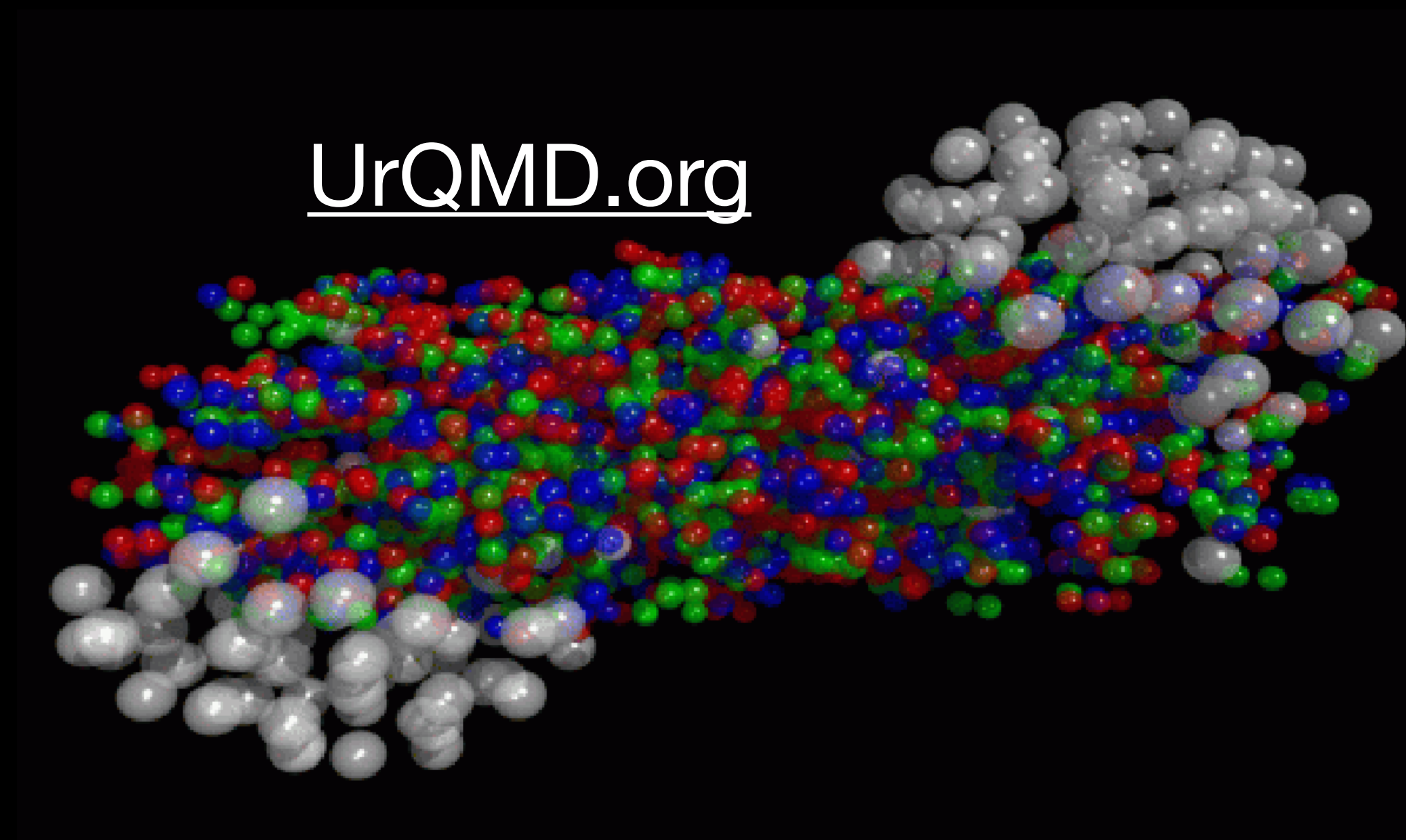
Initial stage - cold
nuclear matter
properties



$$\rho \approx \rho_0$$

Semi understood

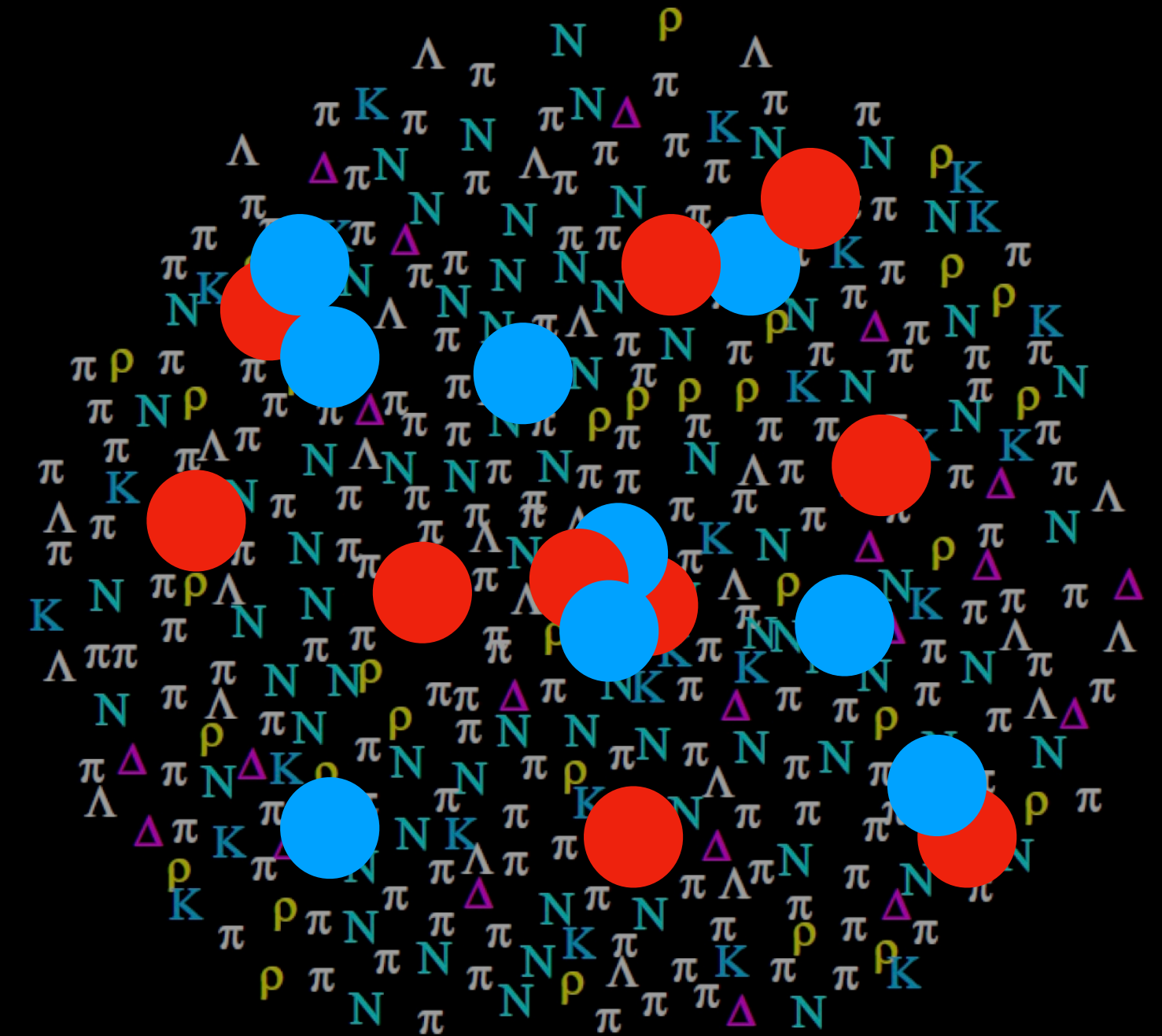
Hot and dense stage
“explosion”



$$\rho > 2\rho_0$$

EoS is here!

Freezing out of
interactions and detection



$$\rho < \rho_0$$

© J. Cleymans

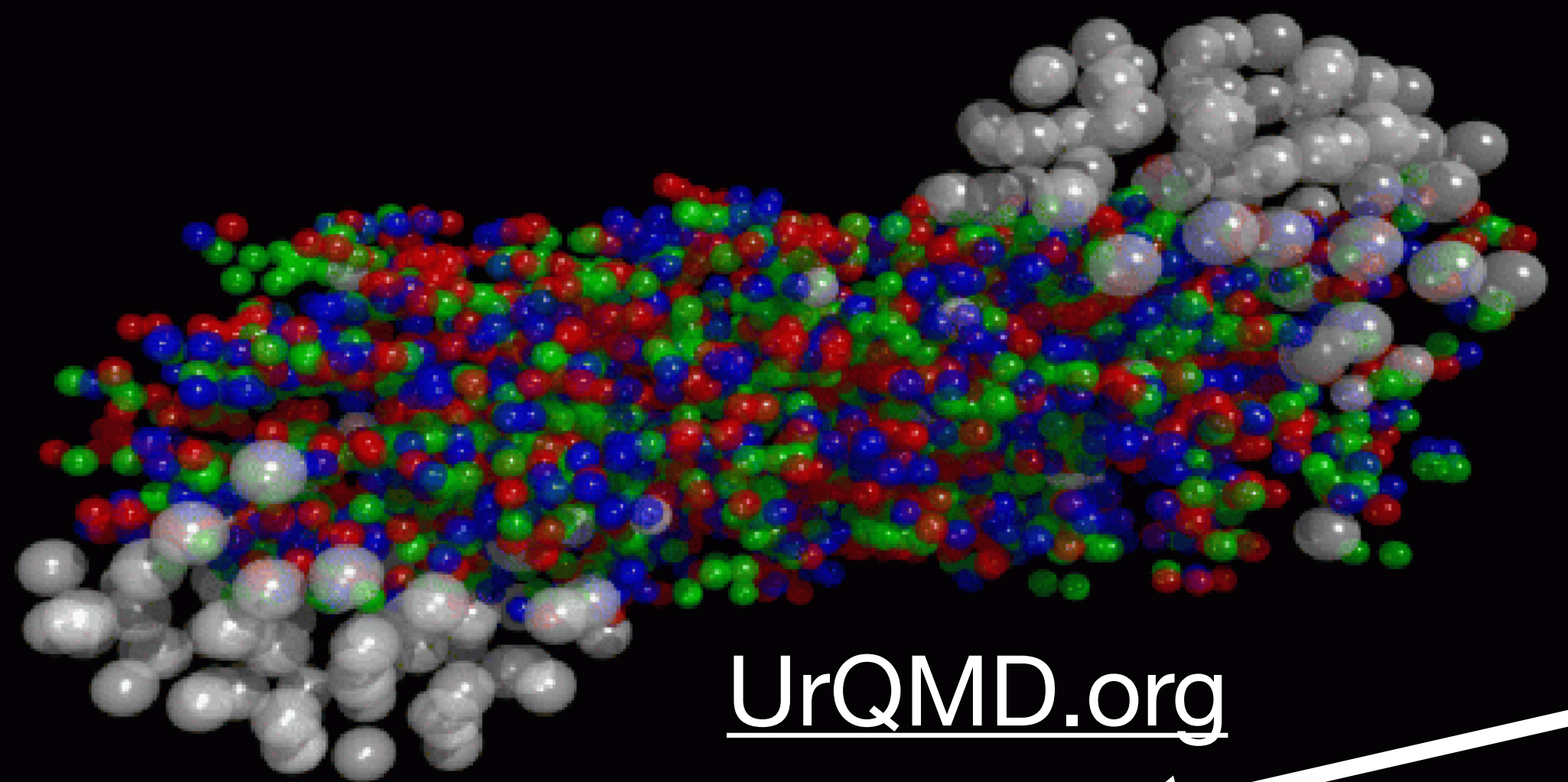
Boring(?) gas

Nuclear fragments abundant - challenge for transport description!

Transport models

[Prog.Part.Nucl.Phys. 134 (2024) 104080]

5

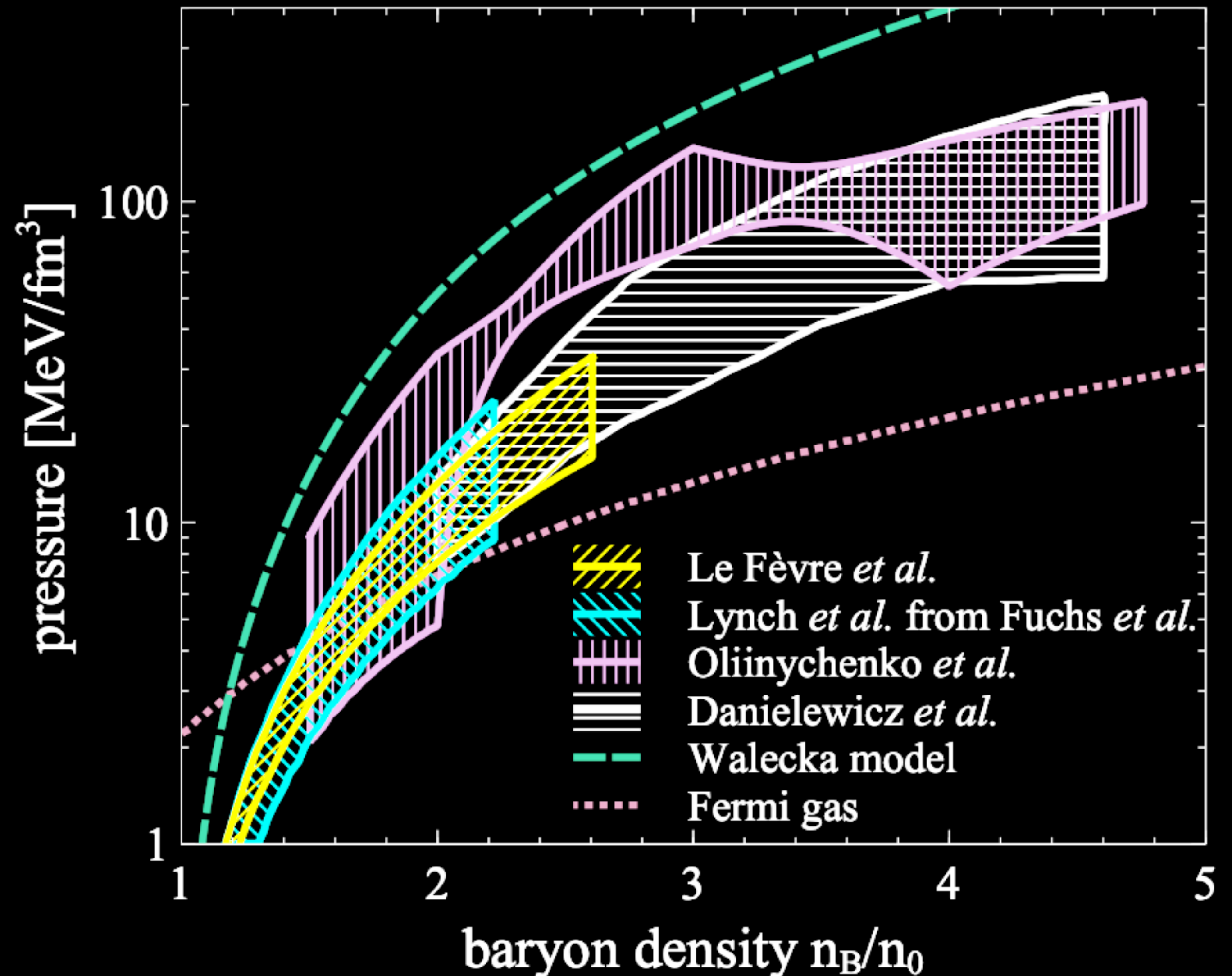


$$\dot{r} = \frac{\partial \langle H \rangle}{\partial p} \quad \dot{p} = - \frac{\partial V(n_B(r))}{\partial n_B(r)} \frac{\partial n_B(r)}{\partial r}$$

$$n_B(r) = \sum_j \left(\frac{\alpha}{\pi} \right)^{3/2} B_j \exp \left(-\alpha (r - r_j)^2 \right)$$

Propagate wave-packets in density dependent potentials between stochastic scattering

How do deal with clusters emerging from collisions?



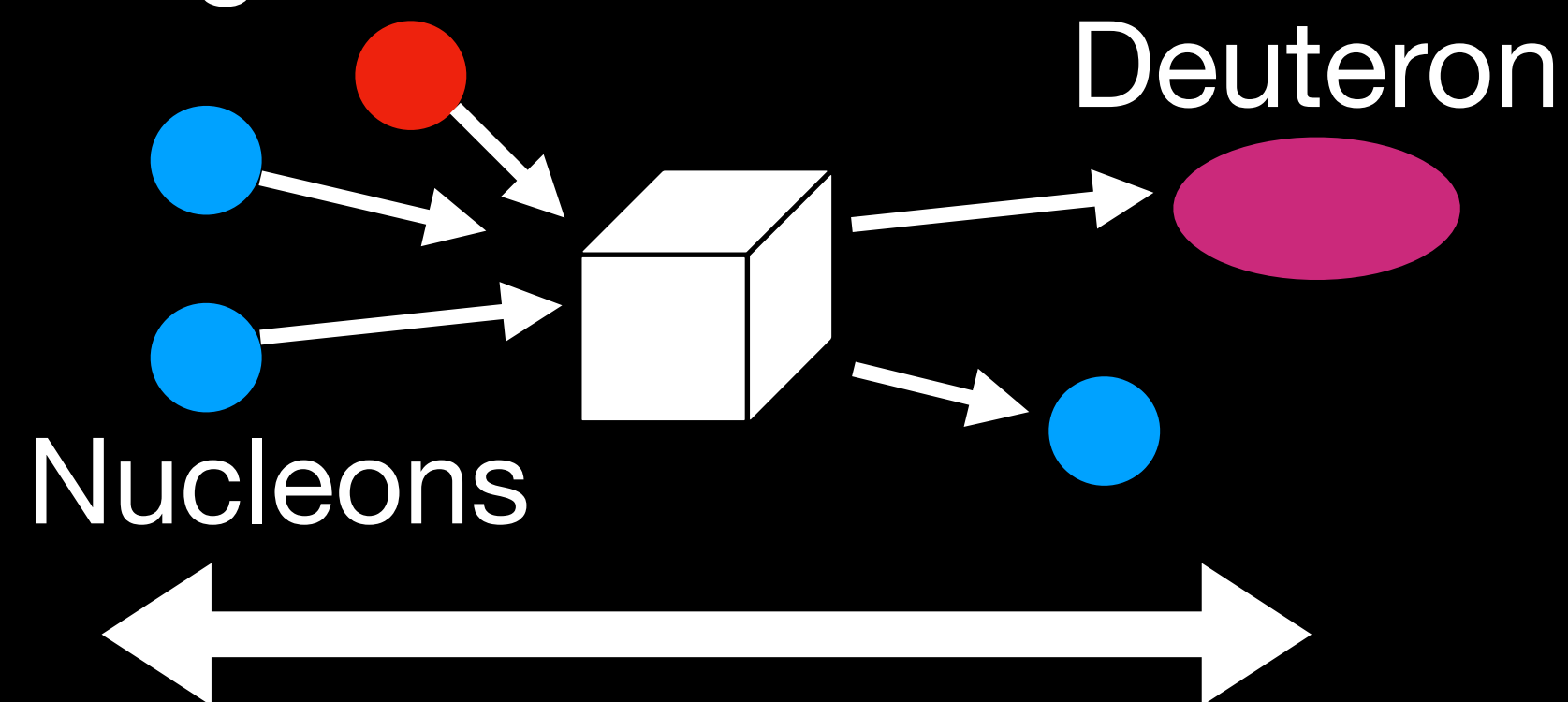
Clustering in transport models

6

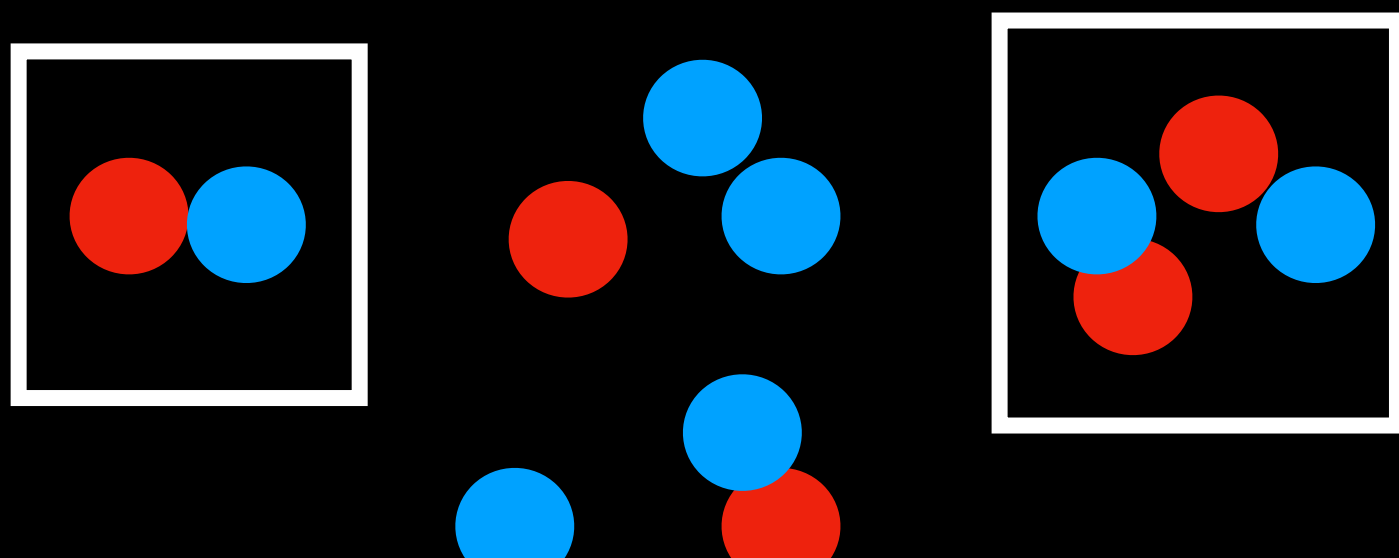
In-medium clustering:

Snowballs in hell

1. Clusters as separate particle species that are propagated alongside nucleons

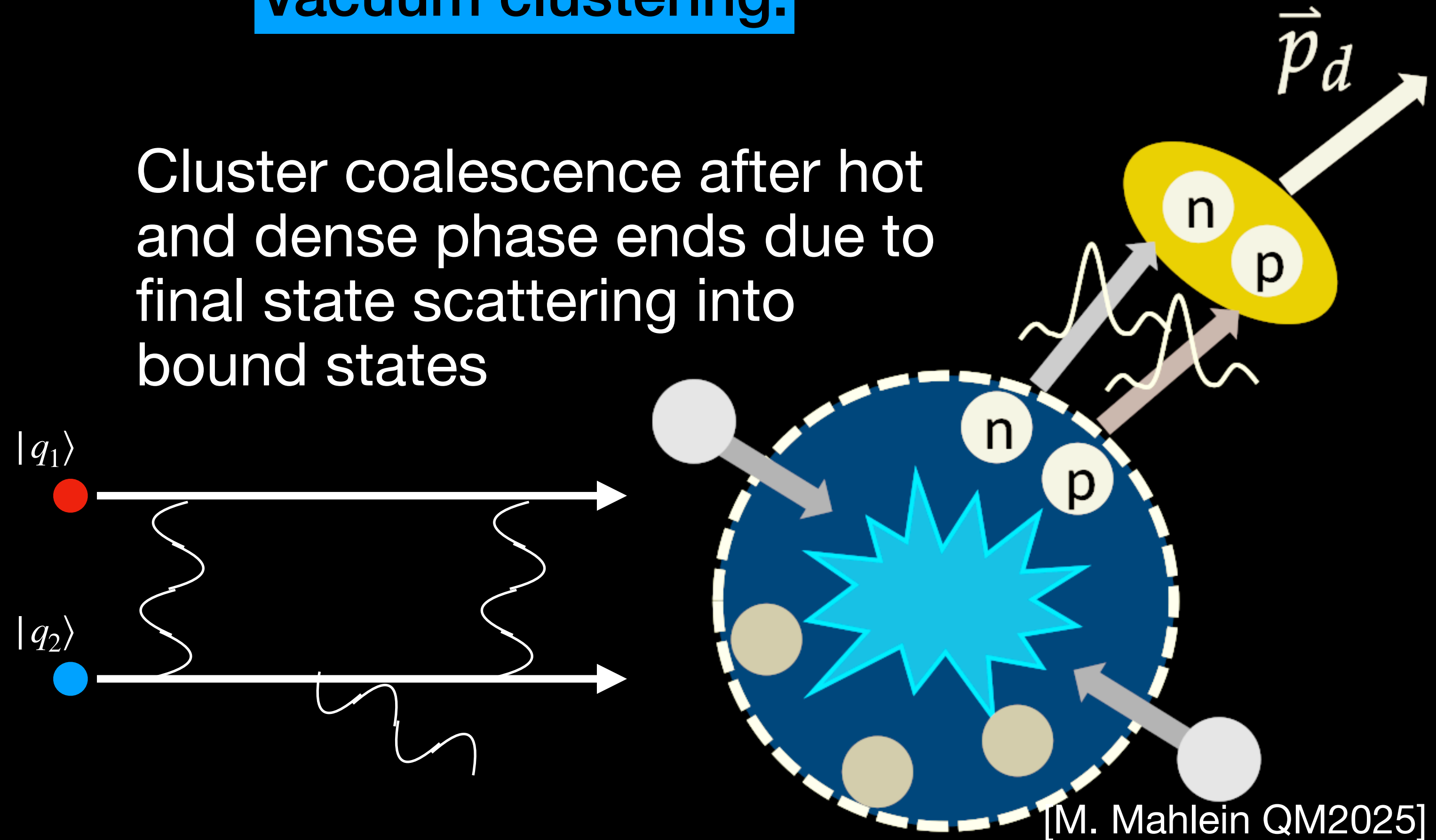


2. Interacting nucleons cluster naturally - cluster identification algorithms are required



Vacuum clustering:

Cluster coalescence after hot and dense phase ends due to final state scattering into bound states



[M. Mahlein QM2025]

Usually very simplified but still have a lot of parameters that require fine tuning

Can we simplify the problem?

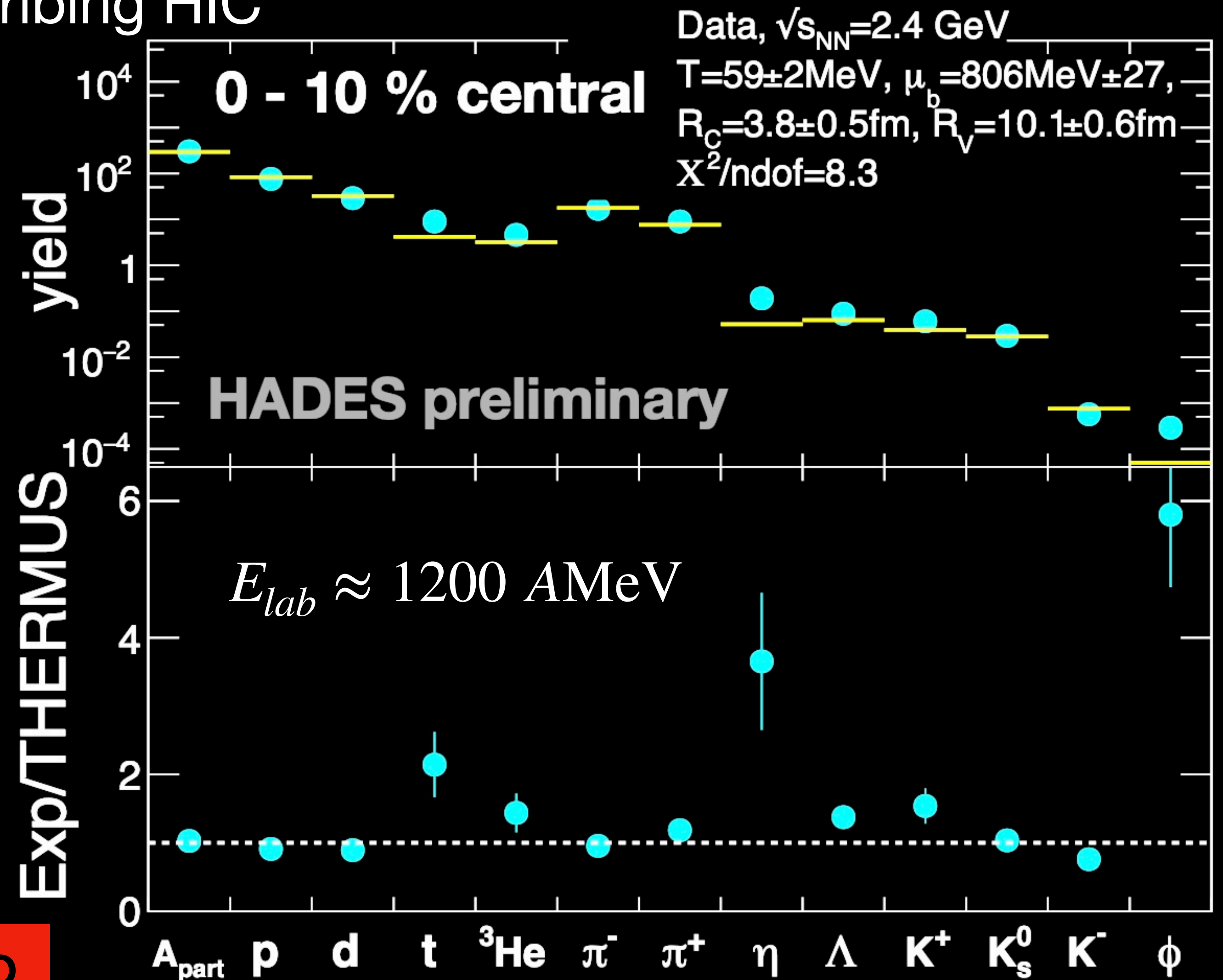
Thermal model success

Equilibrium is very successful in describing HIC

In particular yields of particles can be described assuming a single source with T, μ_B, μ_Q, μ_S

These thermodynamic parameters are usually fitted to the data, but one can extract them from transport linking them to properties of the dense matter

Can one connect thermal model to the matter obtained in the simulation?



Local thermodynamic equilibrium

Instead of whole system at single temperature consider subsystems

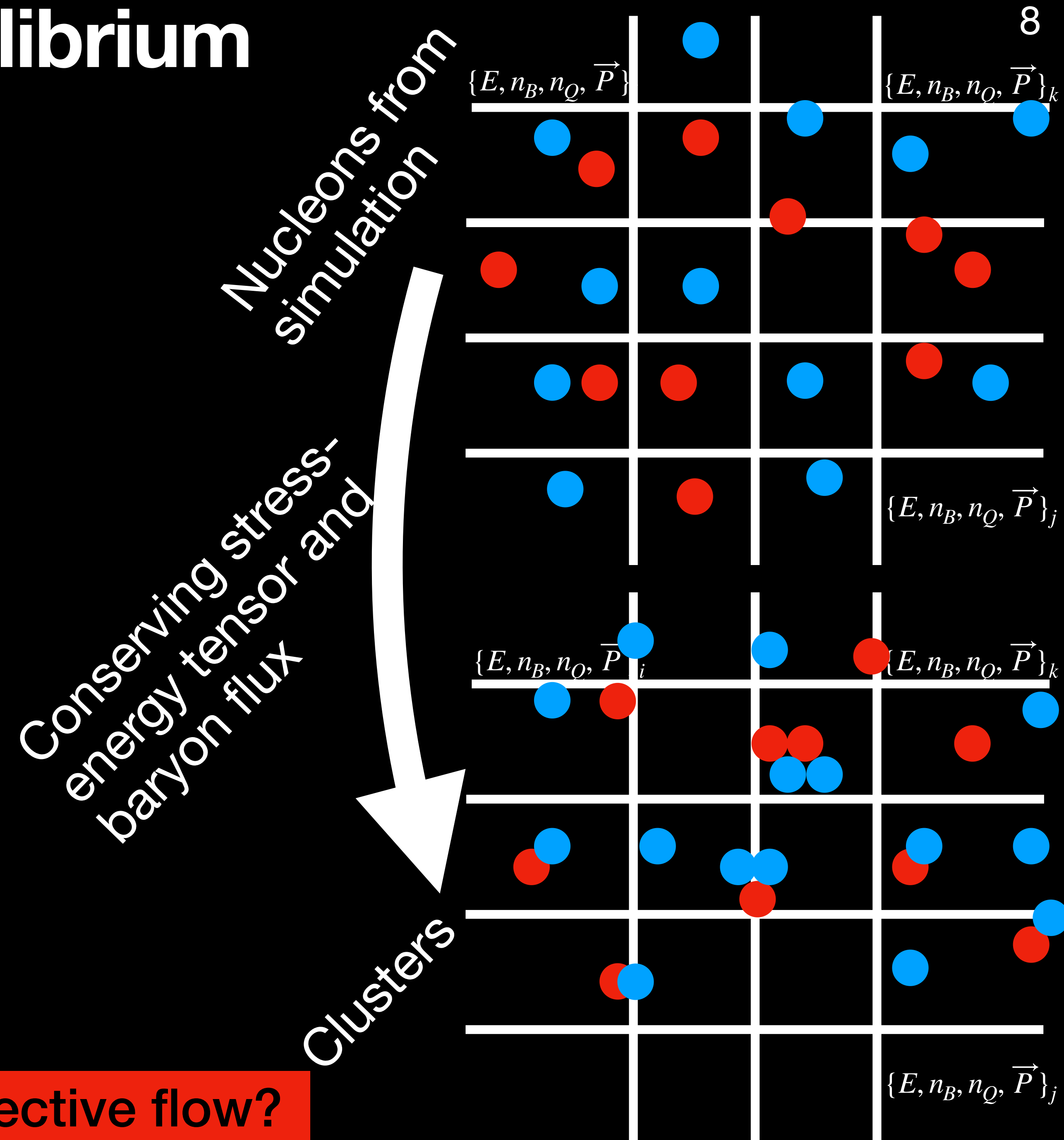
Temperature and density in space-time

Cluster phase space densities defined by conservation laws

Quick and “easy” model for cluster production

Equilibrium is an important baseline for any result

Can be put on top of any transport code (that has conservation laws)



How to separate internal energy from collective flow?

Collective flow and density

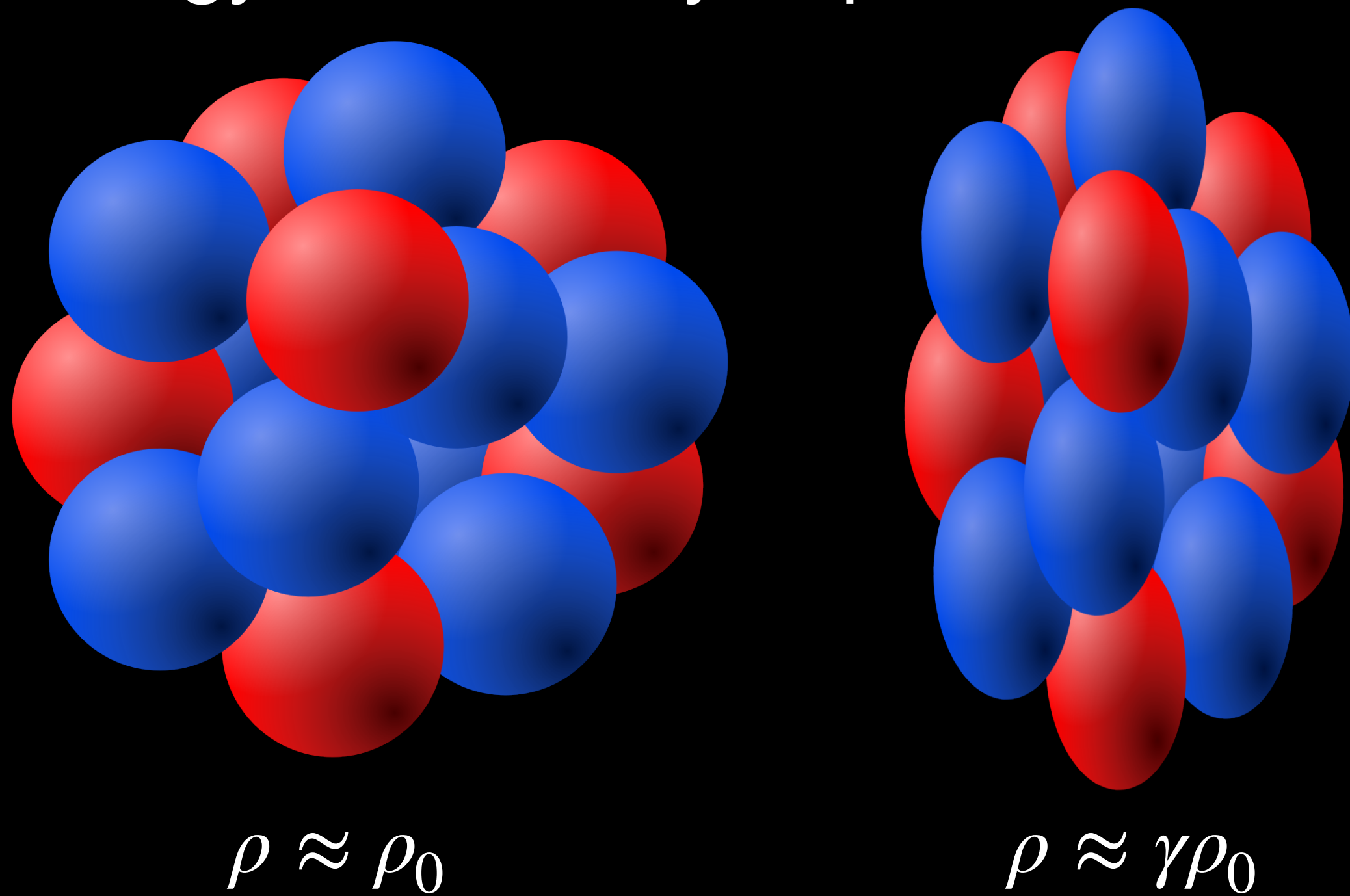
Energy and density depend on a frame

It is natural to consider that nucleon current:

$$j^\mu = \left\langle \sum_i B_i \frac{p^\mu}{p^0} \right\rangle / V$$

With rest frame defined as frame where:

$$j^0 = \rho, \vec{j} = 0, \text{ i.e. } u^\mu = \frac{j^\mu}{\sqrt{j^\mu j_\mu}}$$



But matter essentially the same

Therefore density and energy make sense if defined in a local rest frame

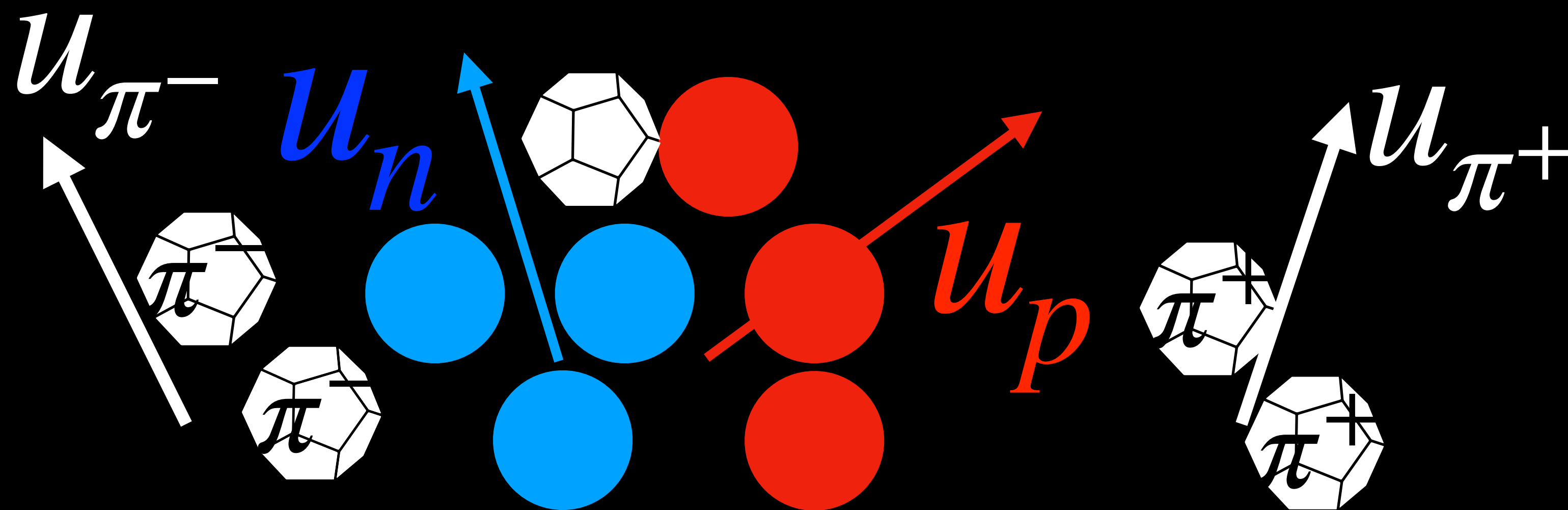


Isospin diffusion

$j_B^\mu = n_B u^\mu$, $n_B = \sqrt{j_B^\mu j_{B,\mu}}$ The Eckart frame associates flow with baryon charge transfer

The isospin transport has to be defined by adding diffusion current: $j_Q^\mu = n_Q u^\mu + j_Q^\mu$

In linear response theory dissipative current proportional to first order gradients of perturbed quantities: $j_Q^\mu = D_Q \Delta^{\mu\nu} \nabla_\nu n_Q$, i.e. one needs to introduce **transport coefficients** to close hydrodynamic system of equations. In transport one can just evaluate all currents and stress energy tensor and then figure out dissipative currents.



Any quantity can have flux associated with it

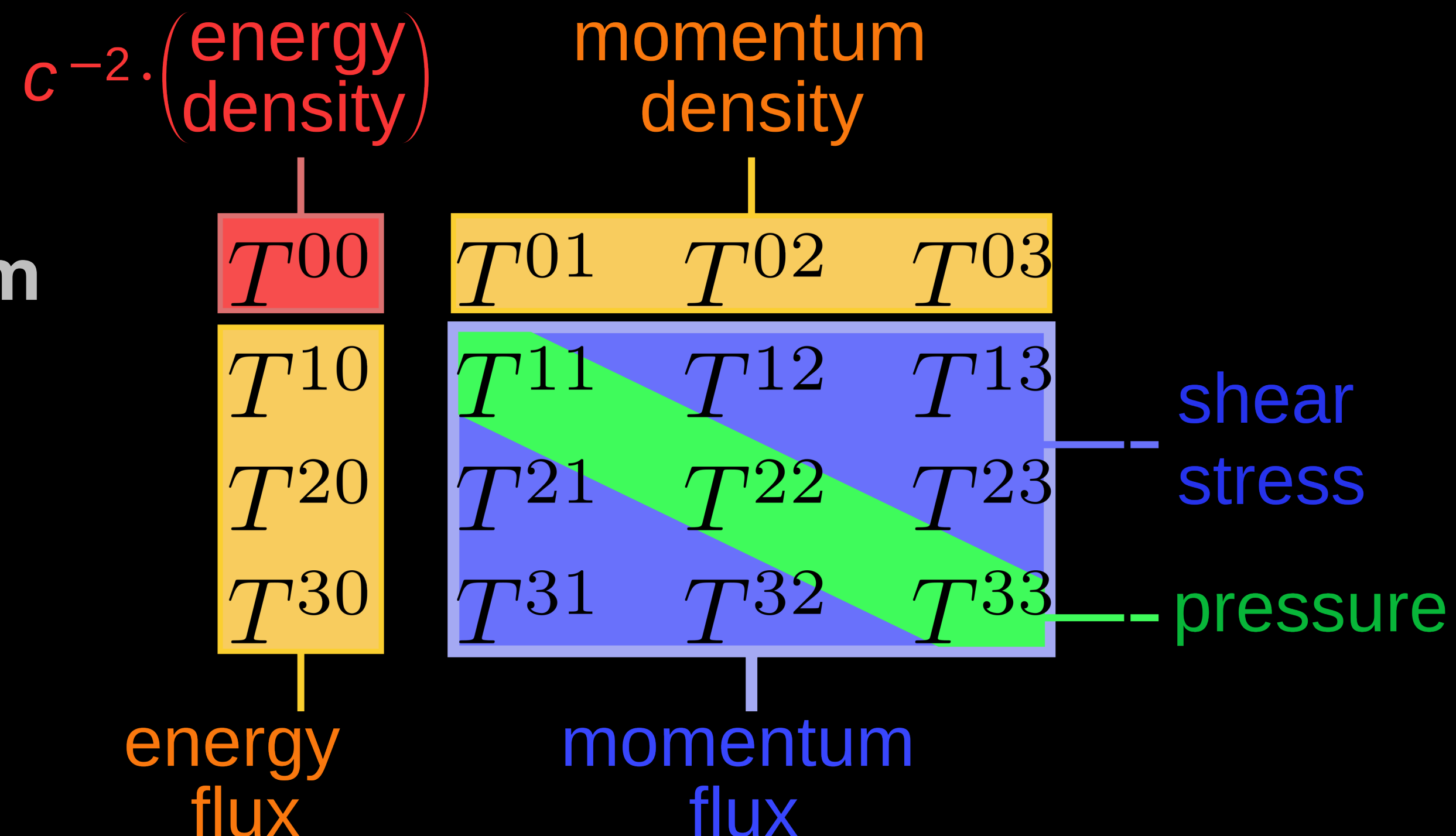
It is believed that fluxes equilibrate quickly (some are 0) and only conserved quantities remain

Equilibrium stress energy tensor

The stress-energy tensor is a symmetric tensor which describes the spacetime density and flux of energy and momentum

$$T^{\mu\nu} = \left\langle \sum_i B_i \frac{p^\nu p^\mu}{p^0} \right\rangle / V + T^{\mu\nu}_{fields}$$

$$T^{\mu\nu}_{EM} = F^{\mu\alpha} F^\nu_{\alpha} - \frac{1}{4} g^{\mu\nu} F^{\alpha\beta} F_{\alpha\beta}$$



In full equilibrium stress energy tensor has to be diagonal in the local rest frame (defined by baryon current):

$$T^{\mu\nu}_{\text{LRF}} = \begin{pmatrix} \epsilon & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{pmatrix}$$

But the tensor extracted from transport will not look like that

Because equilibrium is not ideal and system still relaxes towards it

Stress-energy tensor out of equilibrium

12

$$T^{\mu\nu} = \epsilon u^\mu u^\nu - (P^{eq} + \Pi) \Delta^{\mu\nu} + \pi^{\mu\nu} + q^\mu u^\nu + q^\nu u^\mu$$

$$\Delta^{\mu\nu} = g^{\mu\nu} - u^\mu u^\nu$$

$$\epsilon = u_\mu u_\nu T^{\mu\nu} \quad \text{Energy density}$$

$$q^\mu = \Delta^\mu_\alpha T^{\alpha\beta} u_\beta \quad \text{Heat flux, defines difference in charge and energy transfer}$$

$$P^{eq} + \Pi = -\frac{1}{3} \Delta_{\mu\nu} T^{\mu\nu} \quad \text{Bulk pressure, connected to bulk viscosity and defines how pressure changes when volume changes fast } \Pi = -\zeta \partial_\mu u^\mu,$$

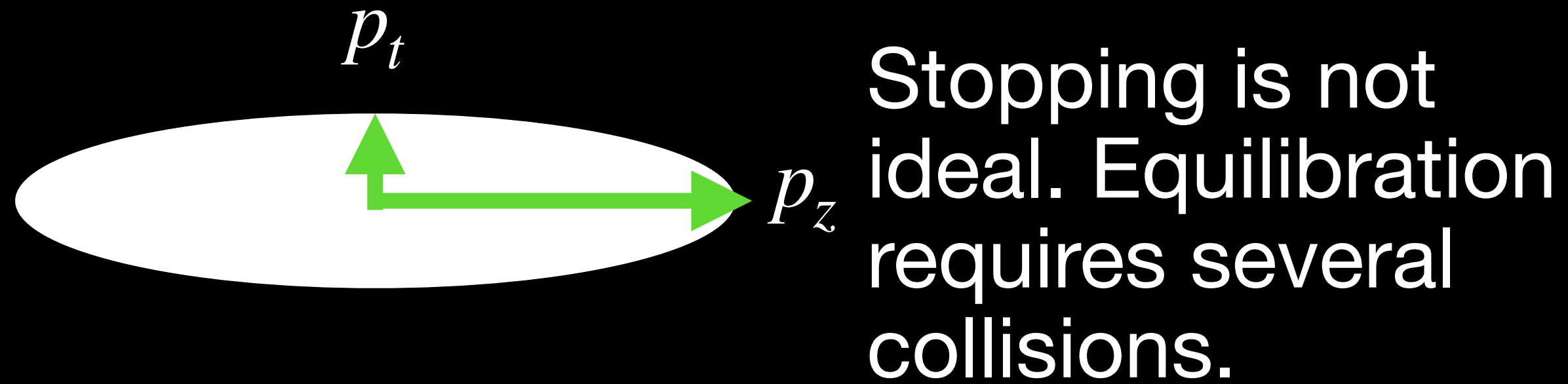
$$\pi^{\mu\nu} = \left(\Delta^\mu_\alpha \Delta^\nu_\beta - \frac{1}{3} \Delta^{\mu\nu} \Delta_{\alpha\beta} \right) T^{\alpha\beta} \quad \text{Shear viscosity, Navier-Stokes term}$$

Distribution function has to match $T^{\mu\nu}$:

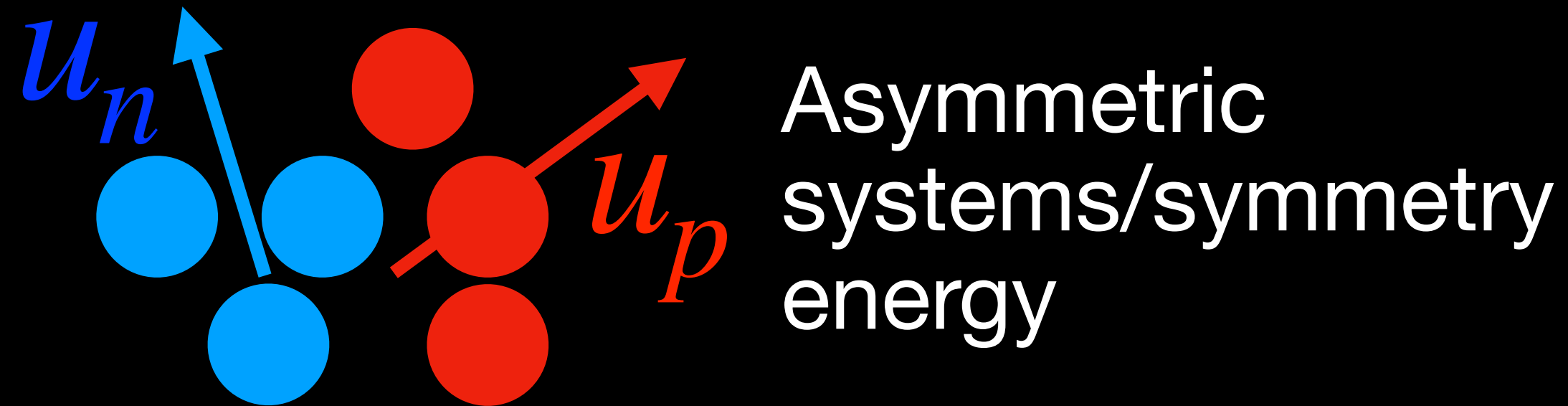
$$T^{\mu\nu} = \int \frac{d^3 p}{p_0} f(t, x, p) p^\mu p^\nu$$

Common non-equilibrium effects

1. Persistent of longitudinal motion



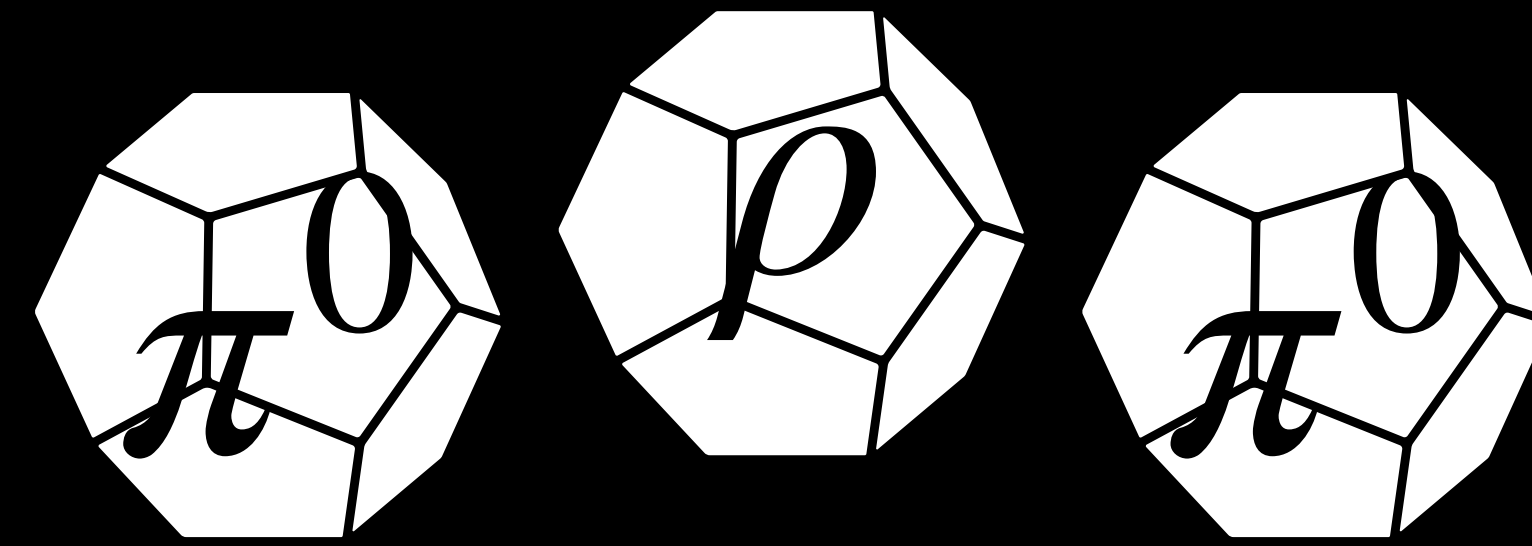
2. Isospin diffusion



3. Chemical non-equilibrium

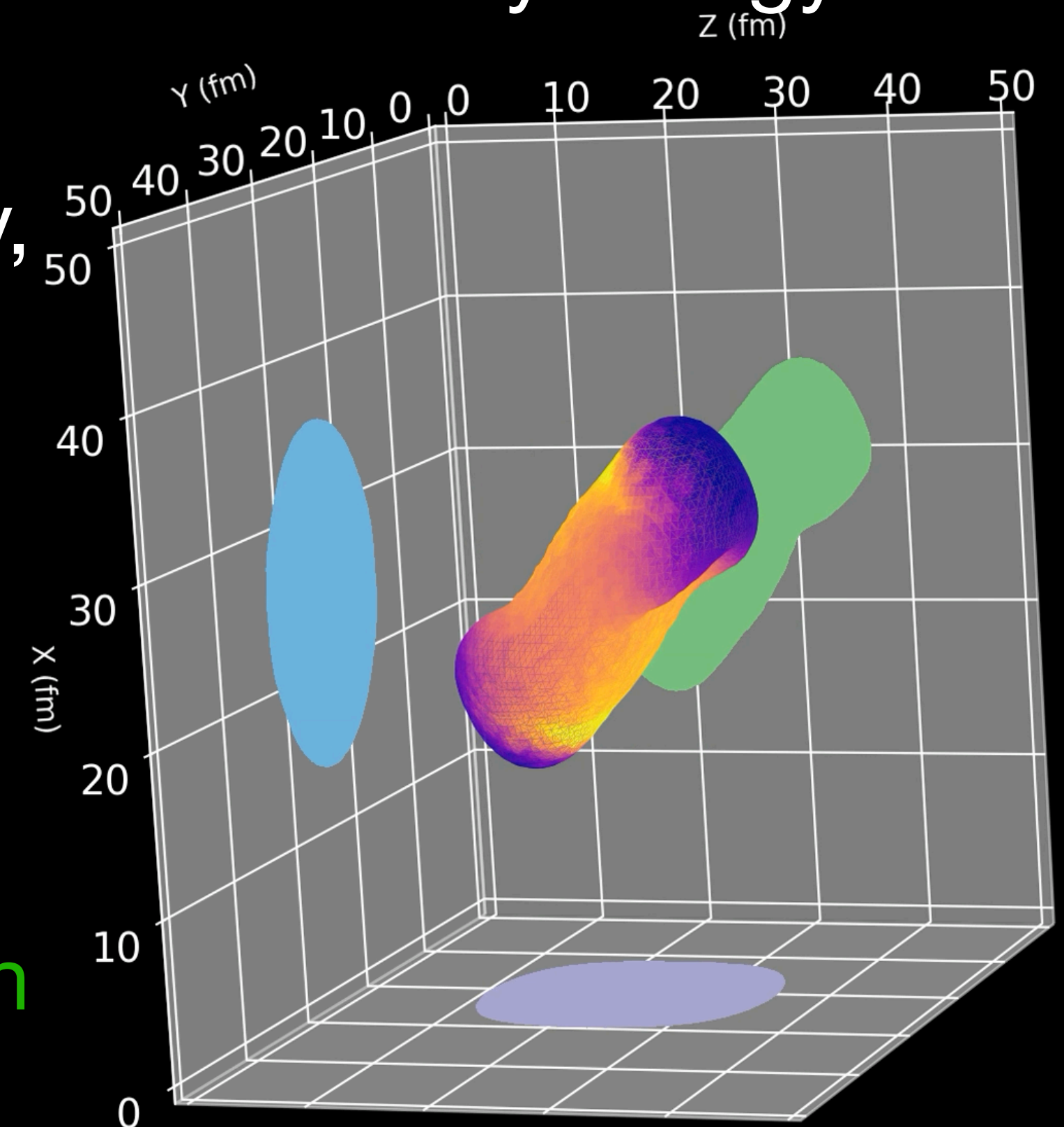


4. Pions/energy flux



Gradients of energy density, charge

Strong initially all these effects weaken towards freeze-out



Local energy density

Advectional transport corresponds to subtraction of translational energy $\mathcal{E} = T^{00} - T^{0i}v_i = (E - \vec{P}\vec{v})/V$

Ignores dissipation, e.g. heat conduction - small effect in rapid collision

Transport: Gas at equilibrium:

$$\langle \varepsilon_{\text{tr}} \rangle(r_{\text{fr}}, t_{\text{fr}}) = \varepsilon_{\text{IG}}[T, \mu_B, \mu_{I_3}, \mu_S](r_{\text{fr}}, t_{\text{fr}})$$

$$\langle n_{\text{B,tr}} \rangle(r_{\text{fr}}, t_{\text{fr}}) = n_{\text{B,IG}}[T, \mu_B, \mu_{I_3}, \mu_S](r_{\text{fr}}, t_{\text{fr}})$$

$$\langle n_{\text{I,tr}} \rangle(r_{\text{fr}}, t_{\text{fr}}) = n_{\text{I,IG}}[T, \mu_B, \mu_{I_3}, \mu_S](r_{\text{fr}}, t_{\text{fr}})$$

$$\langle n_{\text{S,tr}} \rangle(r_{\text{fr}}, t_{\text{fr}}) = n_{\text{S,IG}}[T, \mu_B, \mu_{I_3}, \mu_S](r_{\text{fr}}, t_{\text{fr}})$$

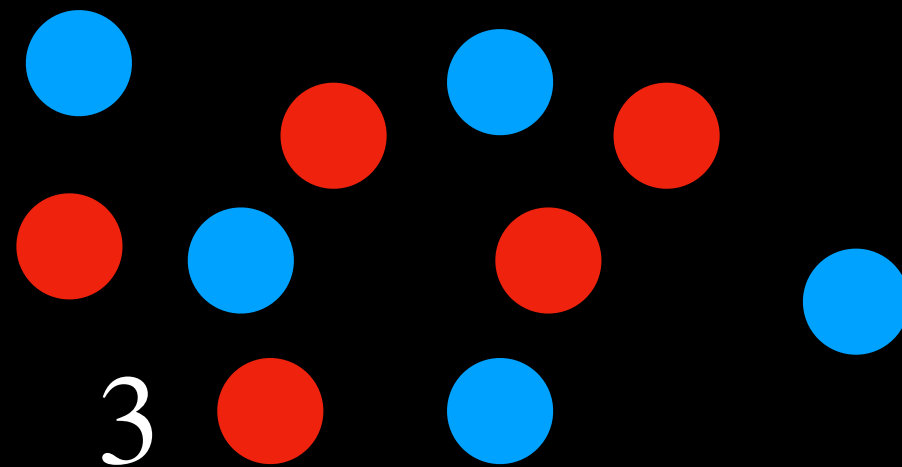
Find temperature and chemical potential

This mapping is not unique

For example one might consider different particle species

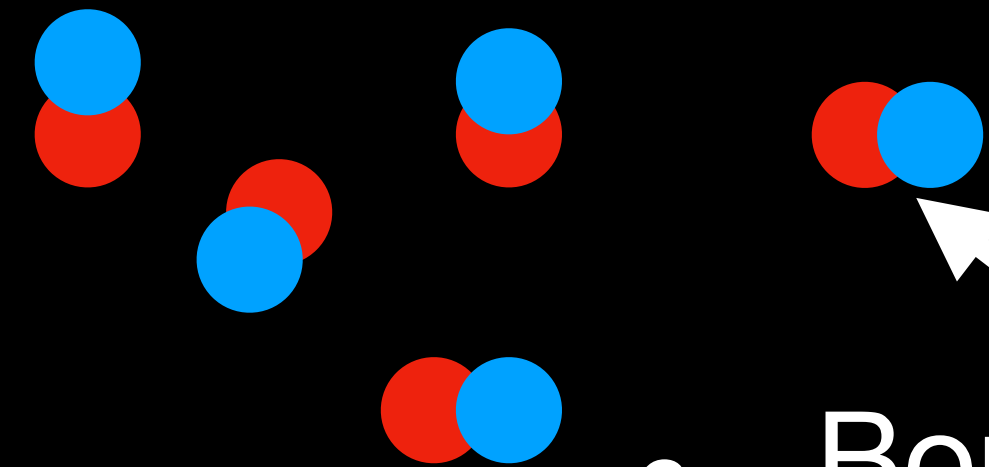
What is the best time to switch?

N nucleons, N particles



$$\mathcal{E} = mN + \frac{3}{2}NT$$

N nucleons, λN particles



$$\mathcal{E} = mN - B(1 - \lambda)N + \frac{3}{2}\lambda NT'$$

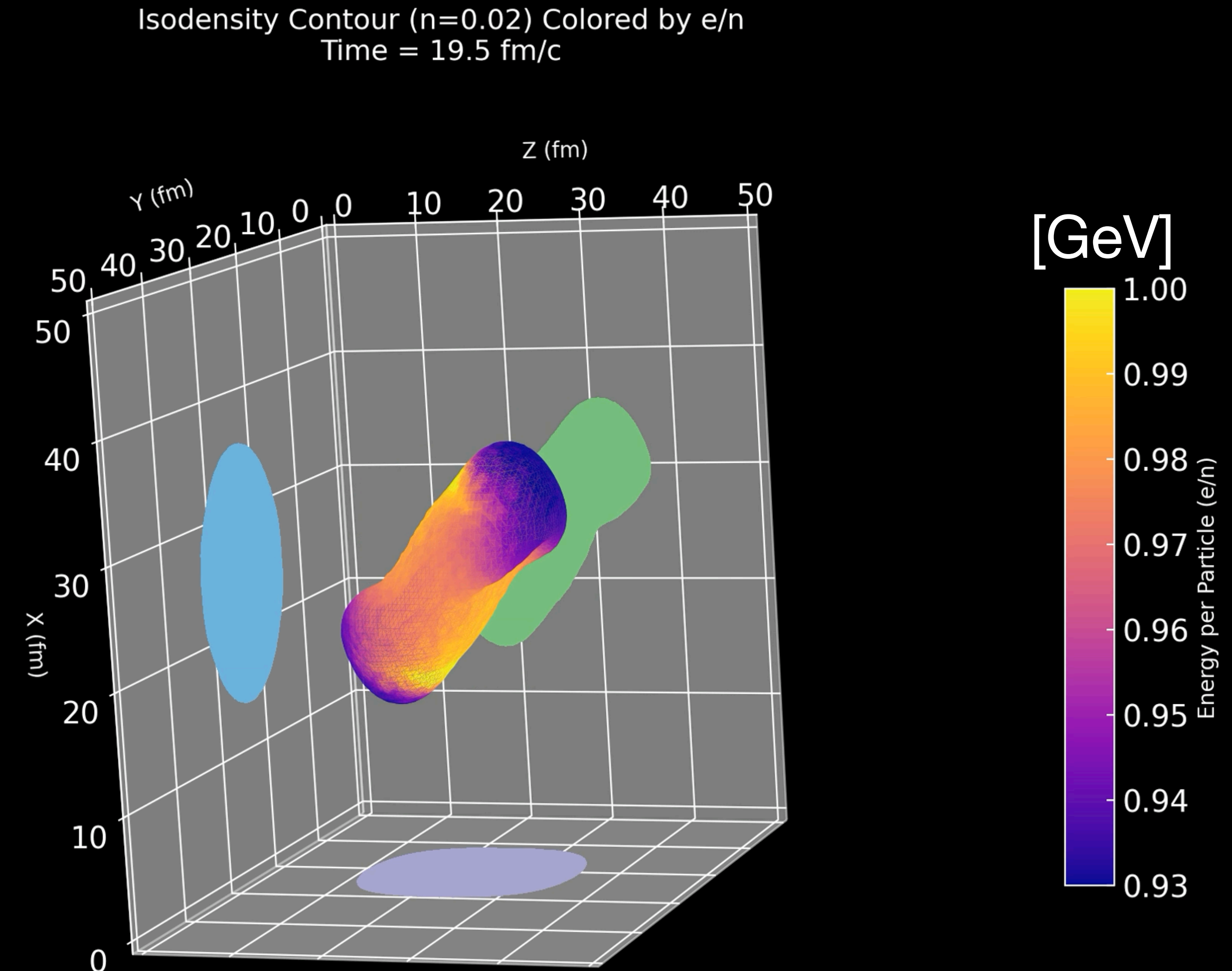
Leads to increase in temperature
Affects free proton observables

Freeze-out hyper surface

If one transitions from nucleons to clusters at a fixed time - mixing of cells with different physical properties occurs.

Instead we choose a transition condition that is motivated by freezing out of interactions

For simplicity we use $n = n_{fr}$



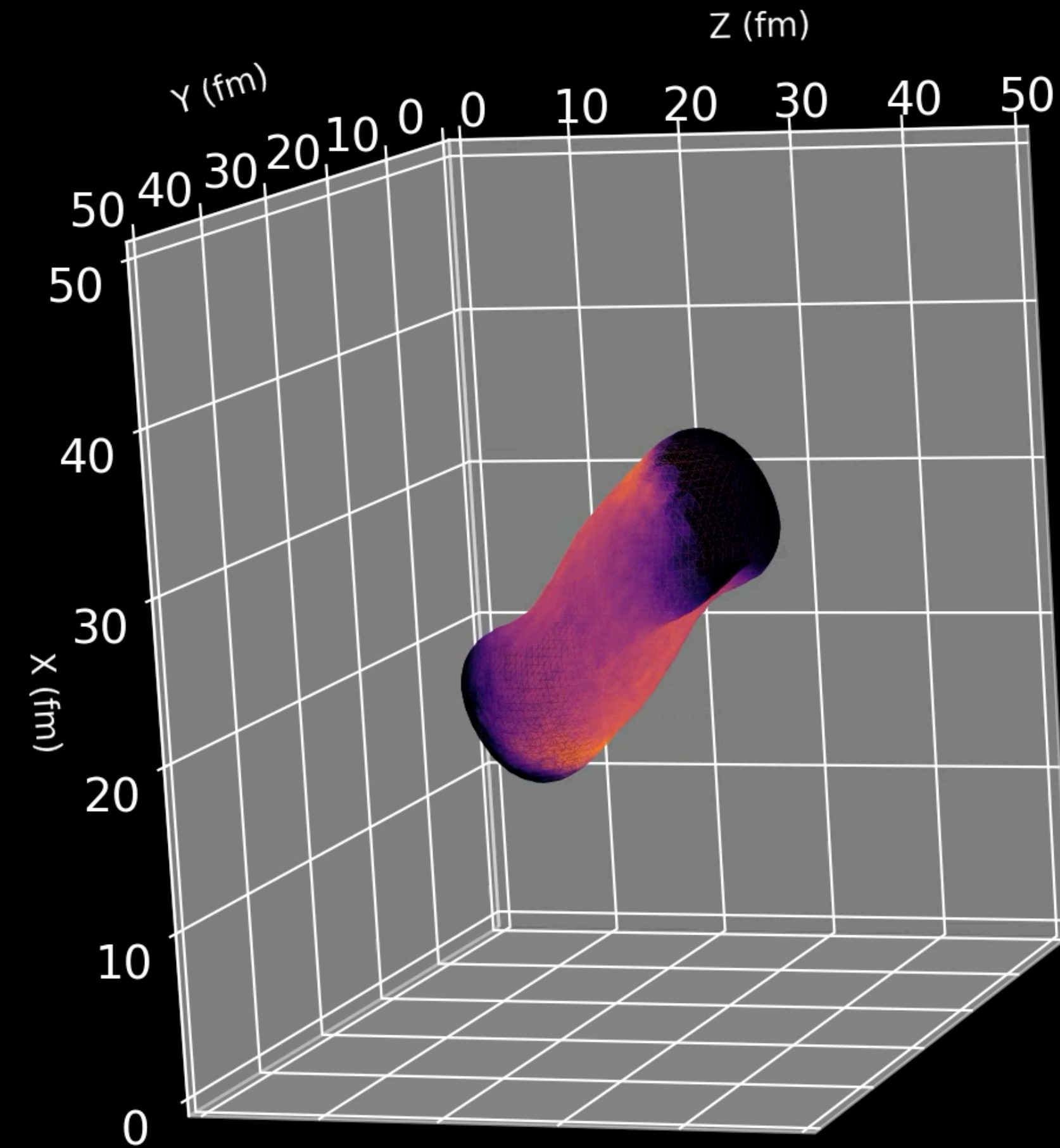
Where is the difference between clusters and nucleons?

Temperature evolution

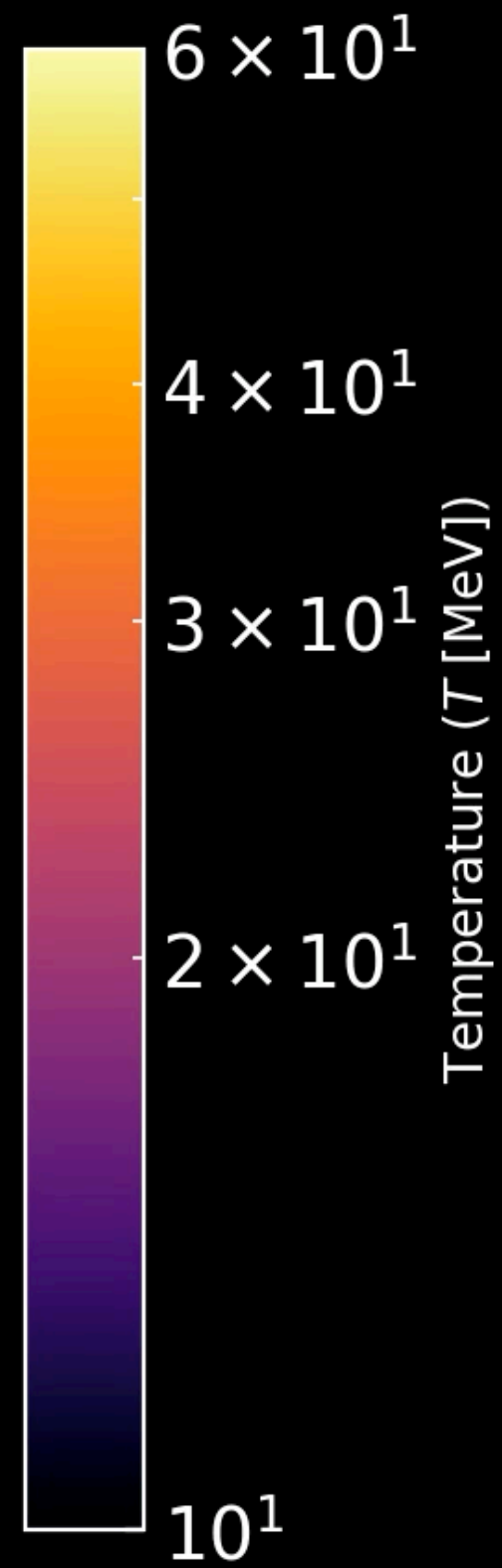
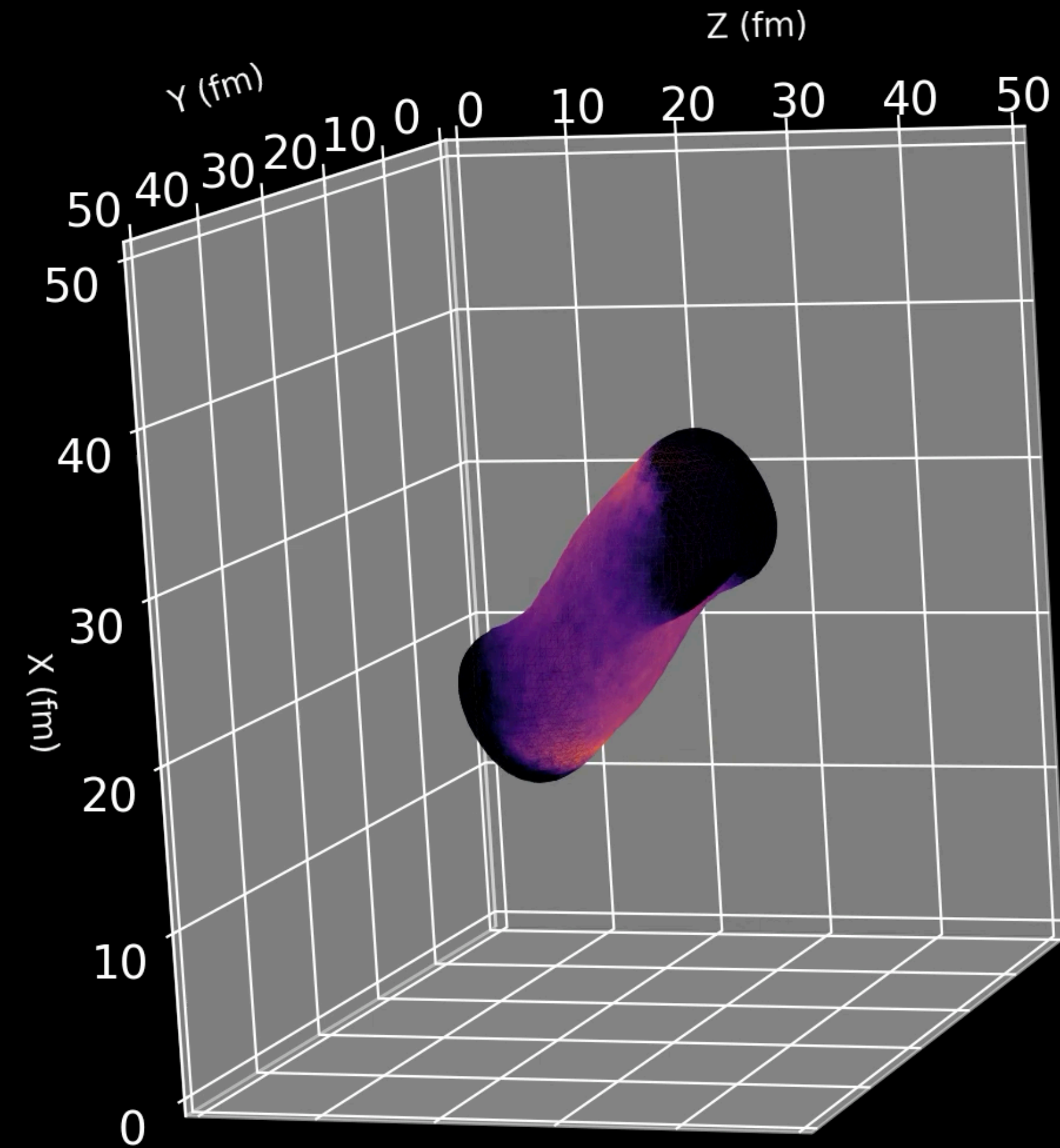
16

Temperature Distribution Comparison | Time = 20.5 fm/c

Clustering



Nucleons



How to get particles from the surface?

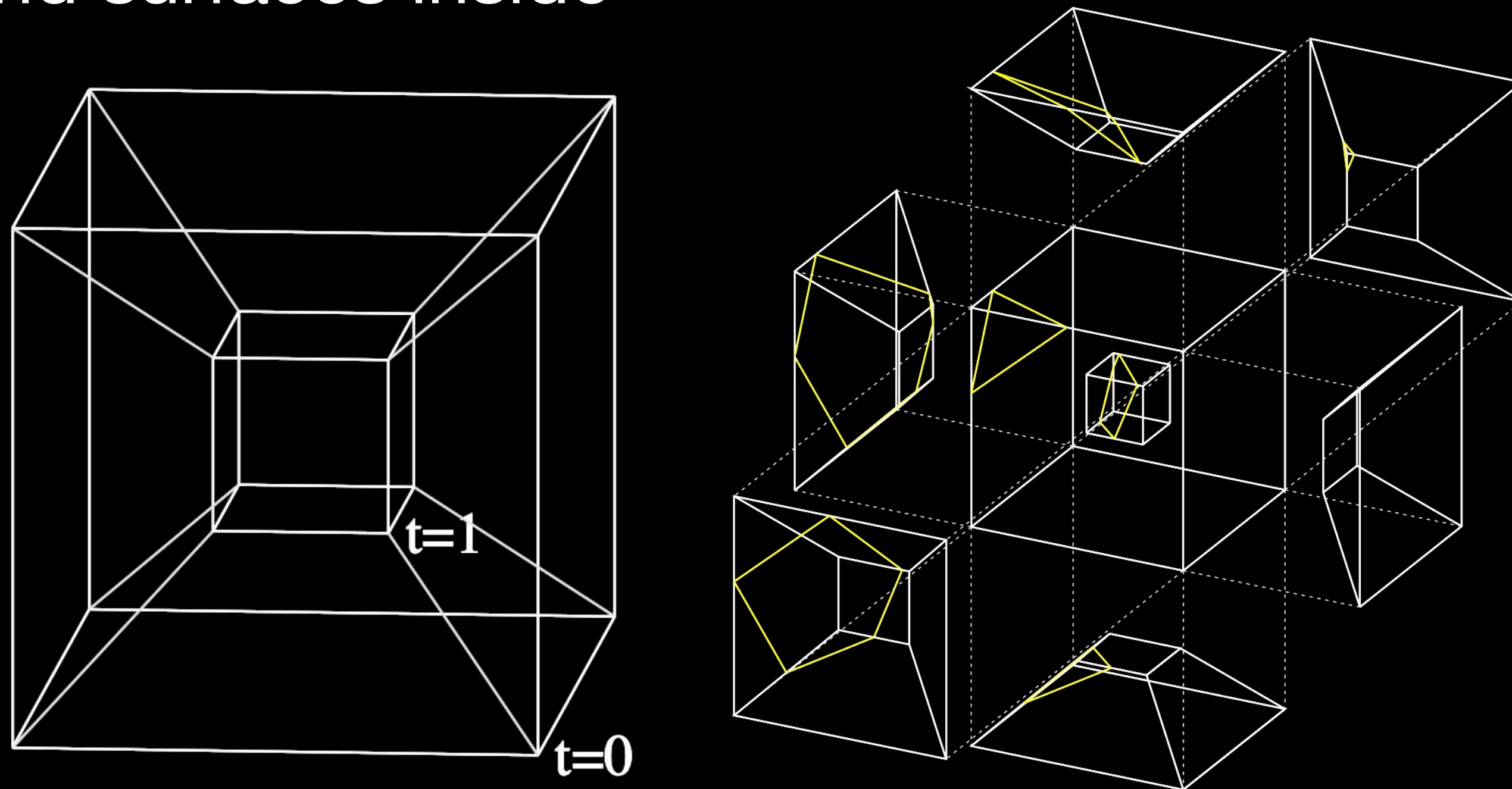
Particle flux

Particles are emitted when shrinking surface crosses them

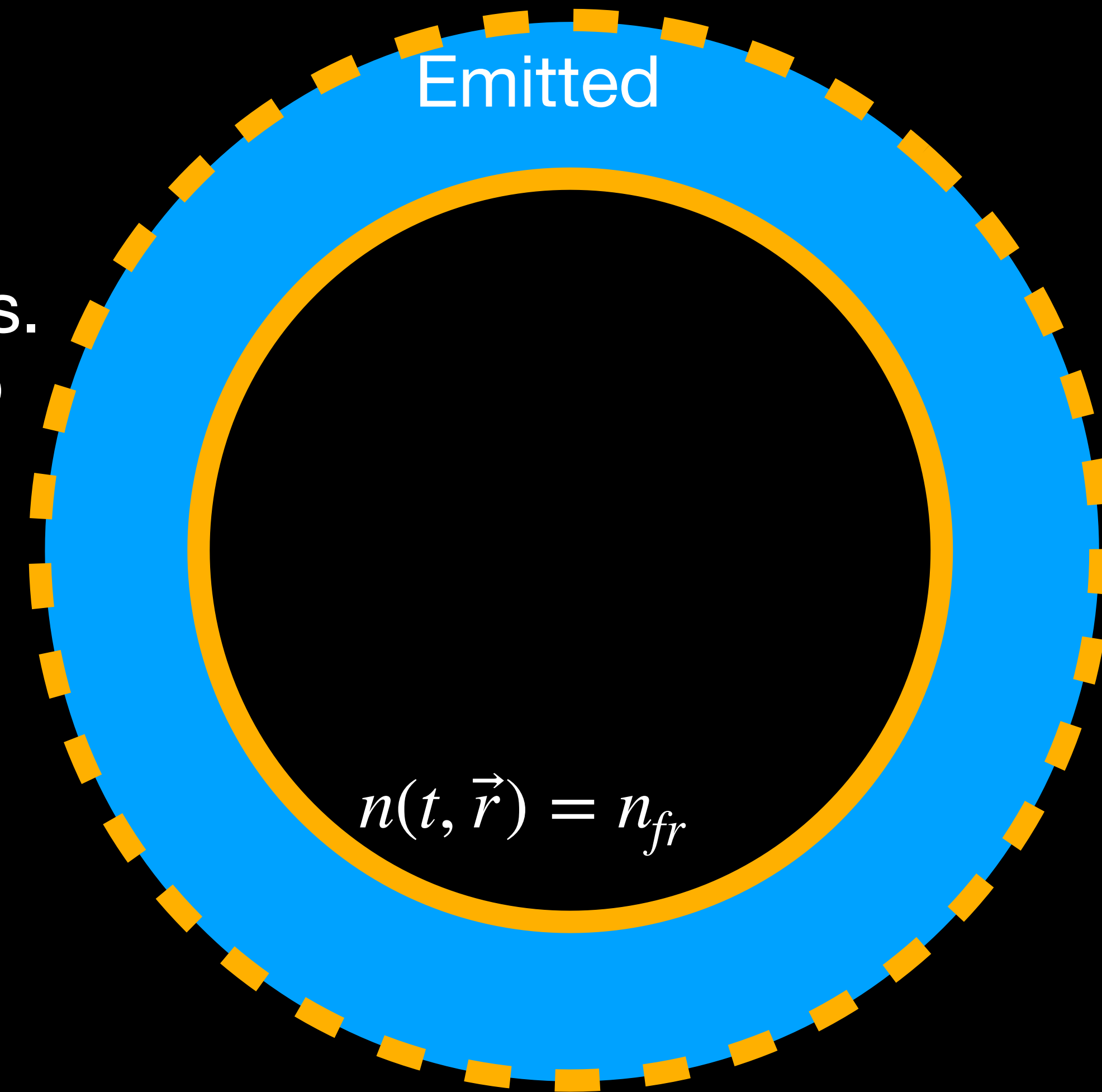
$$d\Sigma_\mu = (dV, \vec{S}dt) \quad p_0 \frac{dN}{d\vec{p}} = p^\mu d\Sigma_\mu f(t, \vec{x}, \vec{p})$$

[F. Cooper and G. Frye, Phys. Rev. D 10, 186 (1974)]

The surface and normals can be treated as 4D objects. Space-time cubes(tesseracts) are then looped over to find surfaces inside



$$n(t_1, \vec{r}) = n_{fr}, t_1 < t$$



[P. Huovinen and H. Peterson, Eur.Phys.J.A 48 (2012) 171]

Phase space densities and partition functions

$$\ln \mathcal{Z} = \sum_i \ln \mathcal{Z}_i$$

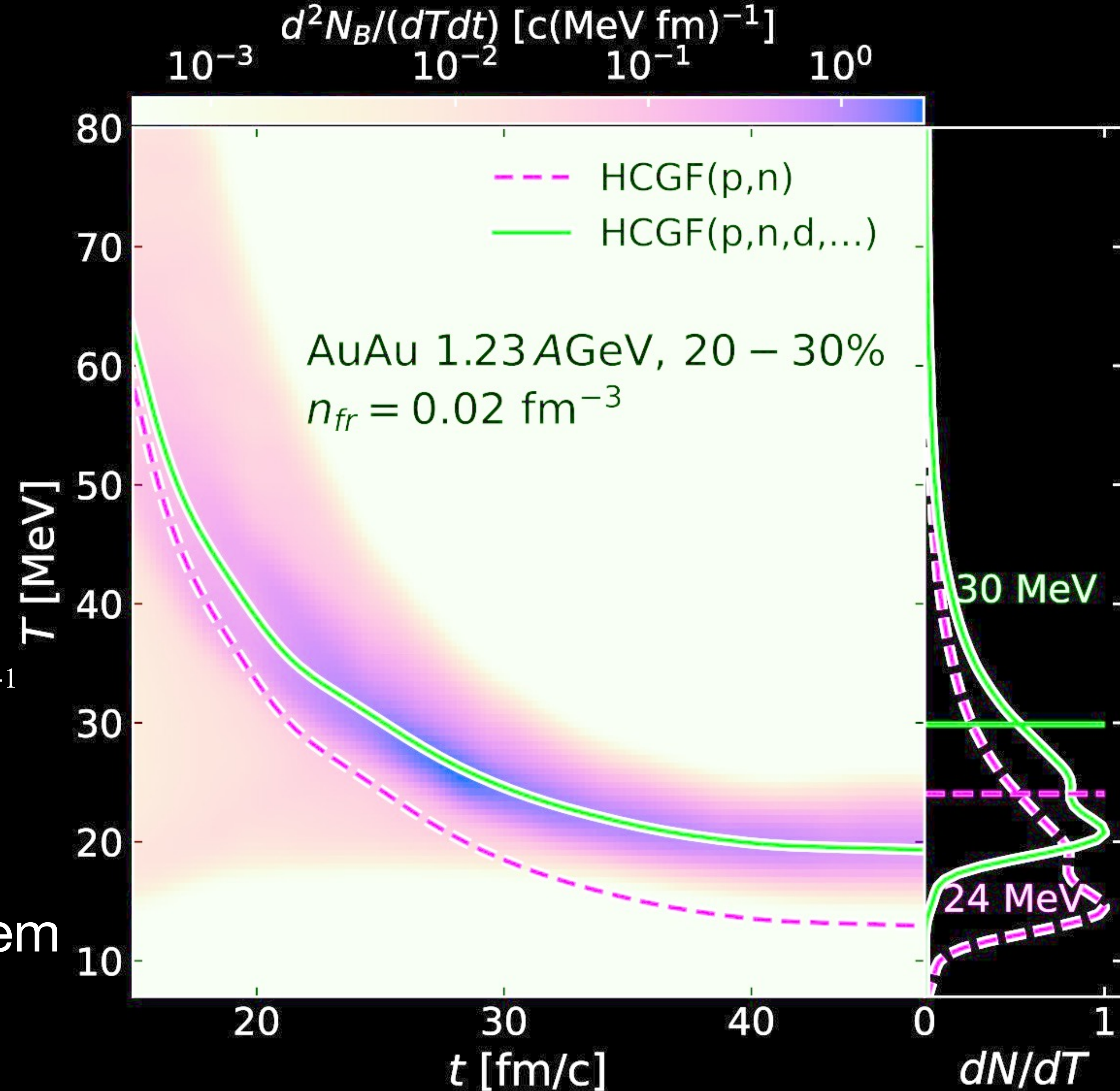
where i runs through nucleons p, n ,
light fragments $d, t, {}^3\text{He}, {}^4\text{He}$

excited states ${}^4\text{He}^*, \dots$ and
unstable isotopes e.g. ${}^5\text{Li}$

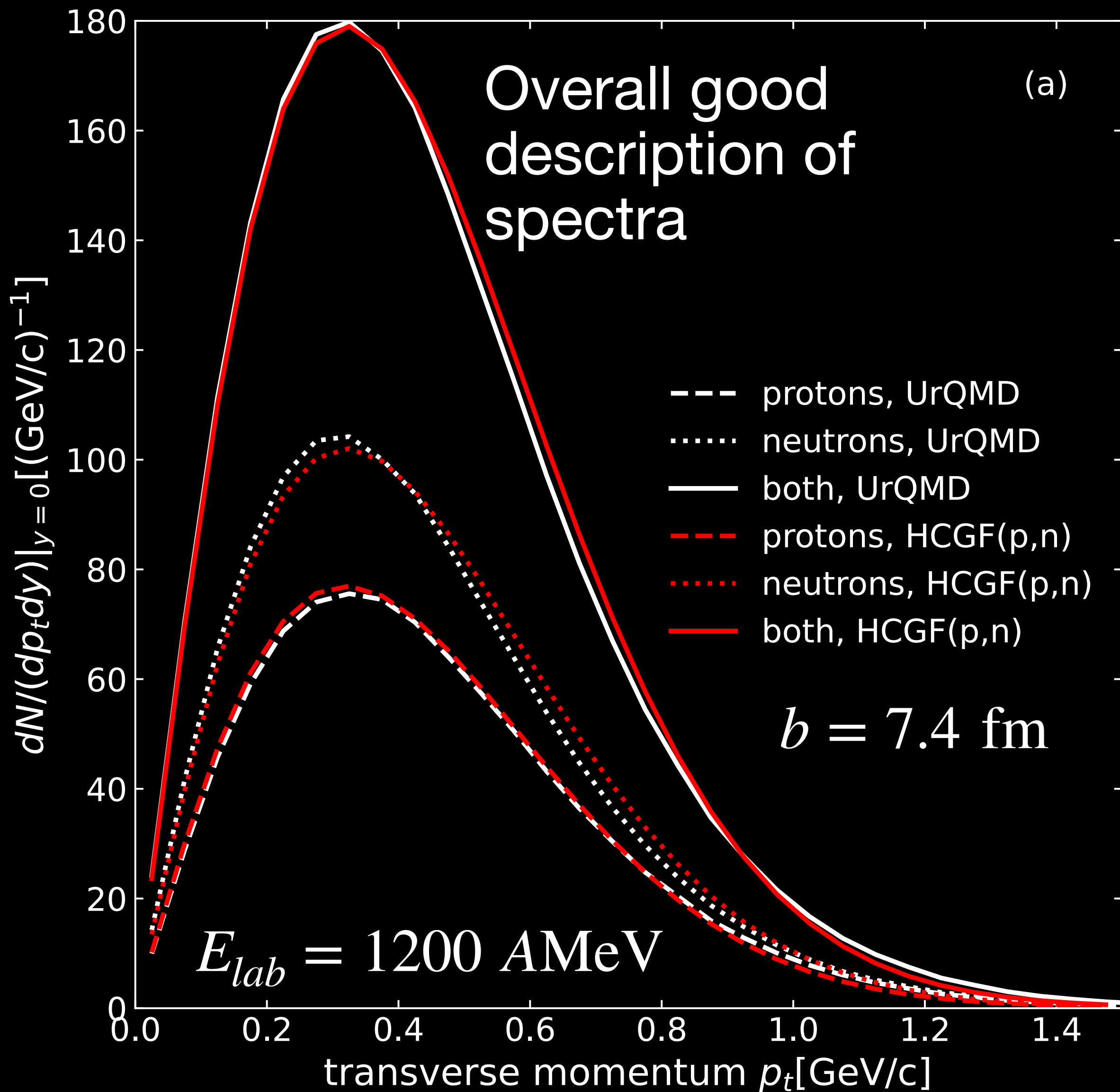
Phase space densities defined by $\vec{v}, T, \mu$

$$f_i(x, p) = \frac{d^6 N_i}{d^3 x d^3 p} = \frac{g_i}{(2\pi)^3} \left[\exp \left(\frac{\gamma(\sqrt{m_i^2 + p^2} - \vec{p} \cdot \vec{v}) - \mu_i}{T} \right) + \eta_i \right]^{-1}$$

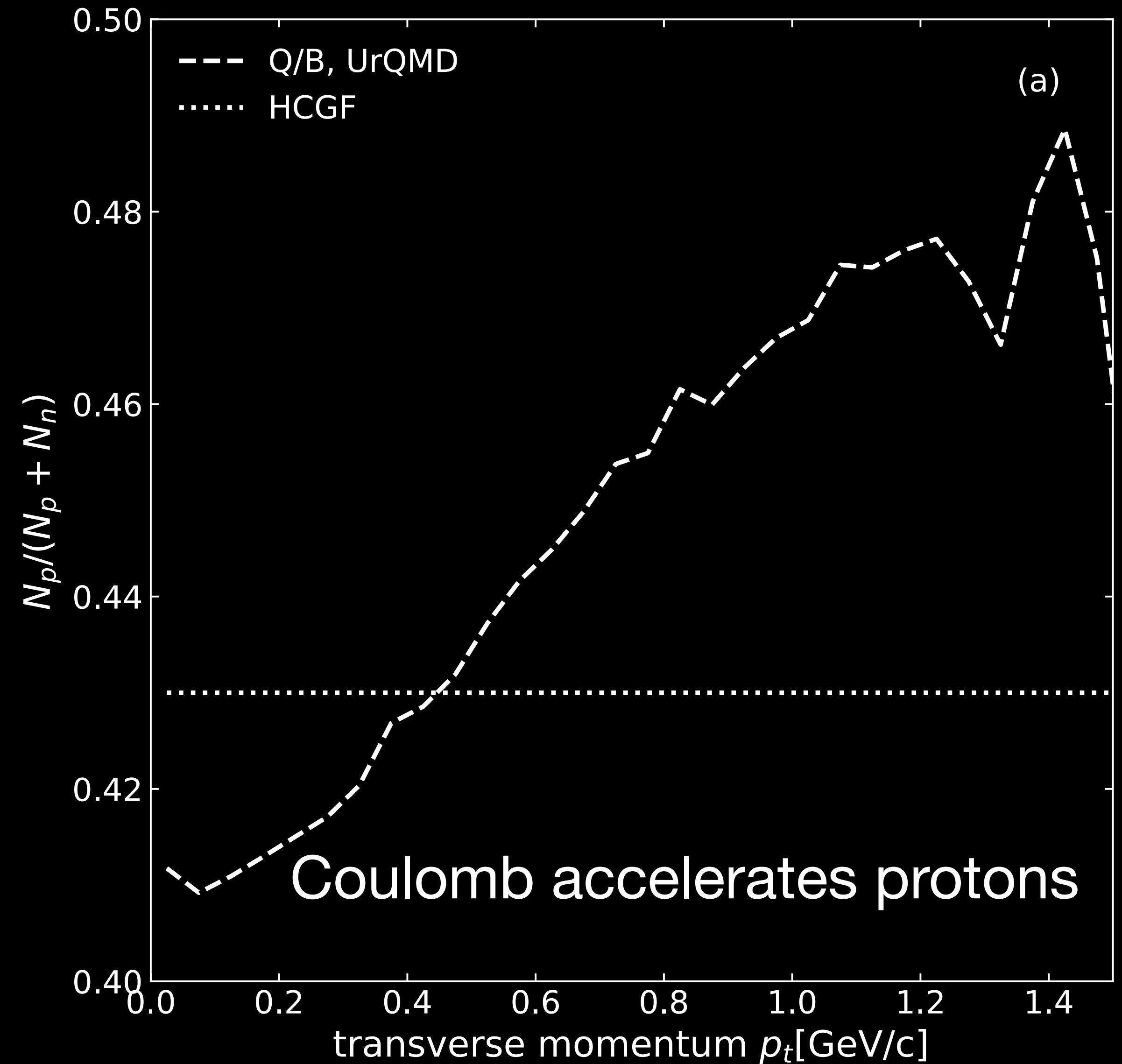
Equilibrium distr. + boost to laboratory system



Conservation law checks



Room for systematic improvement:

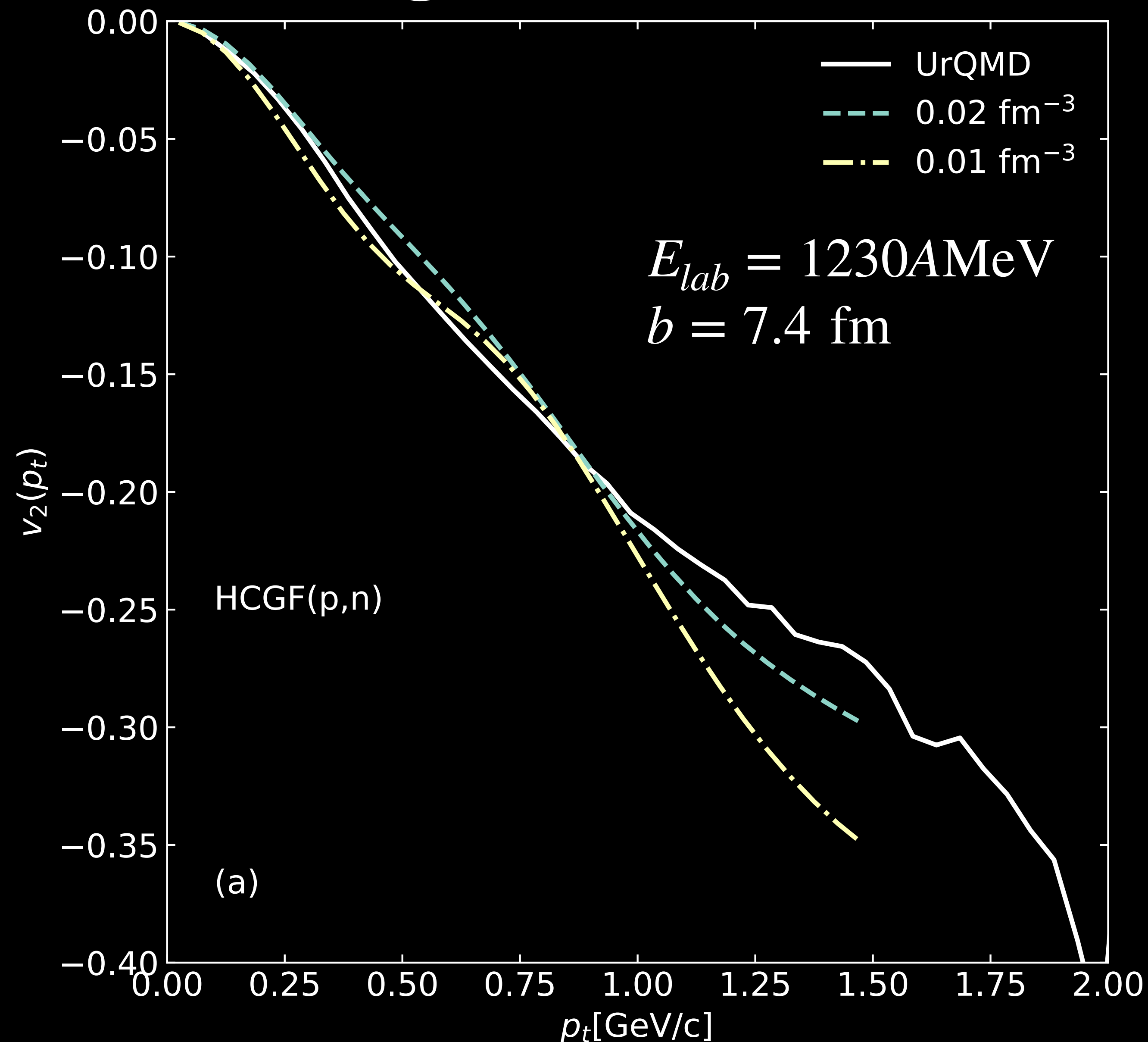


Azimuthal flow before clustering

The freeze-out density is chosen to provide stable results that match the data.

The method appears to be precise enough to describe azimuthal flow.

It is however hard to make decision on freeze-out parameters without considering clustering.



Clustering in thermal model

In non-interacting system yields are driven by the entropy

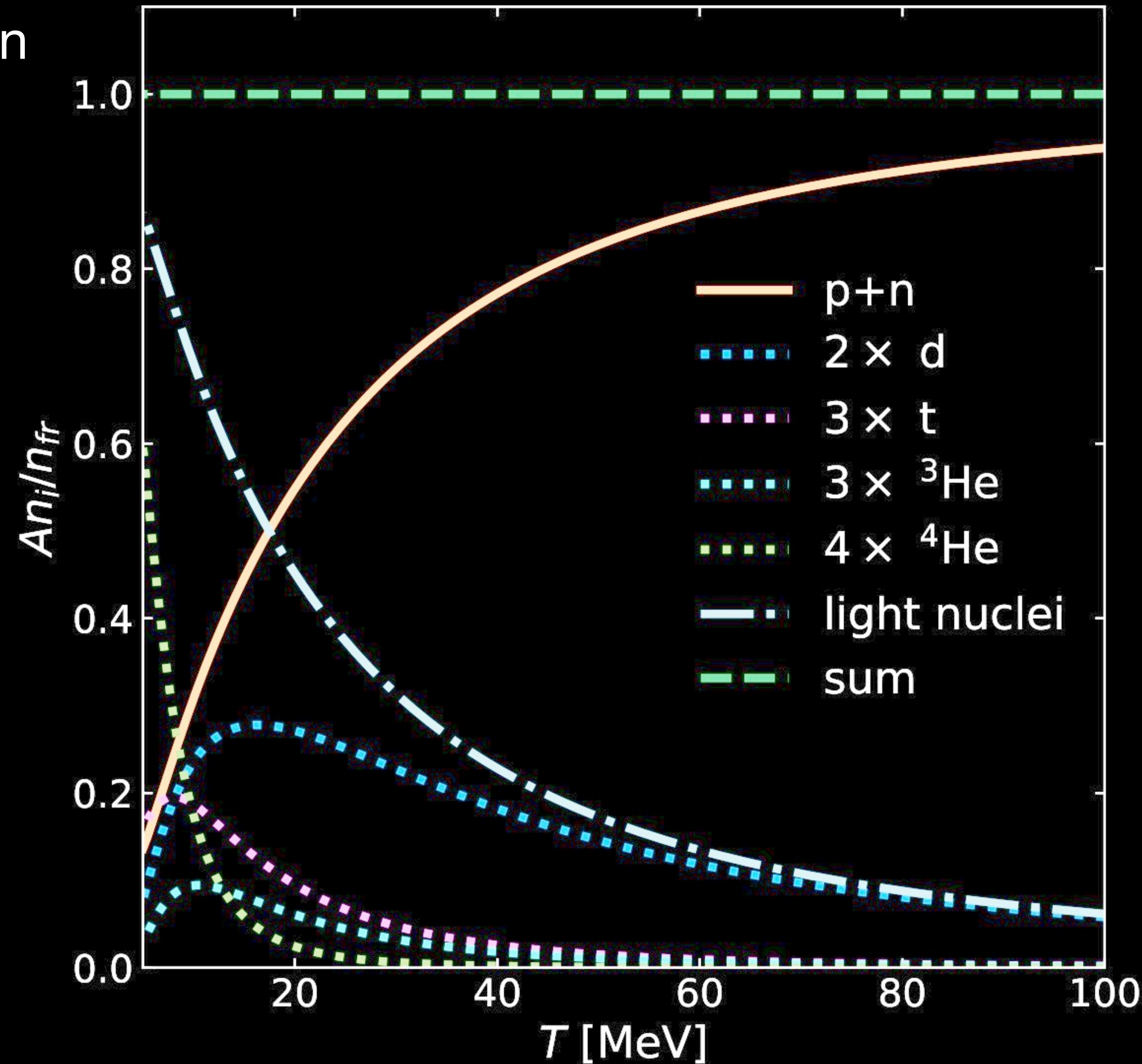
For the same charges system of free nucleons has more degrees of freedom

At high temperature clustering is suppressed

At low temperature thermal excitation in the system are suppressed $e^{-\frac{E}{T}}$

Due to the binding energies clusters can occupy more momentum states - have higher entropy

At the low temperature cluster with the lowest binding energy per particle survives



Clustering in thermal model

To do list

Each nucleon occupies a fraction of momentum space - Mott effects

$$f_{\nu}^{eq} = \frac{H(P - P_{\nu}^{mott})}{\exp^{\frac{e_{\nu}(P) - \mu_{\nu}}{T}} \pm 1}$$

[R.Wang, et al. Phys. Rev. Lett. 136, 172301 (2026)]

[G.Ropke, et al. Phys.Rev.C79:014002,2009]

Each nucleon occupies a fraction of coordinate space - Excluded volume

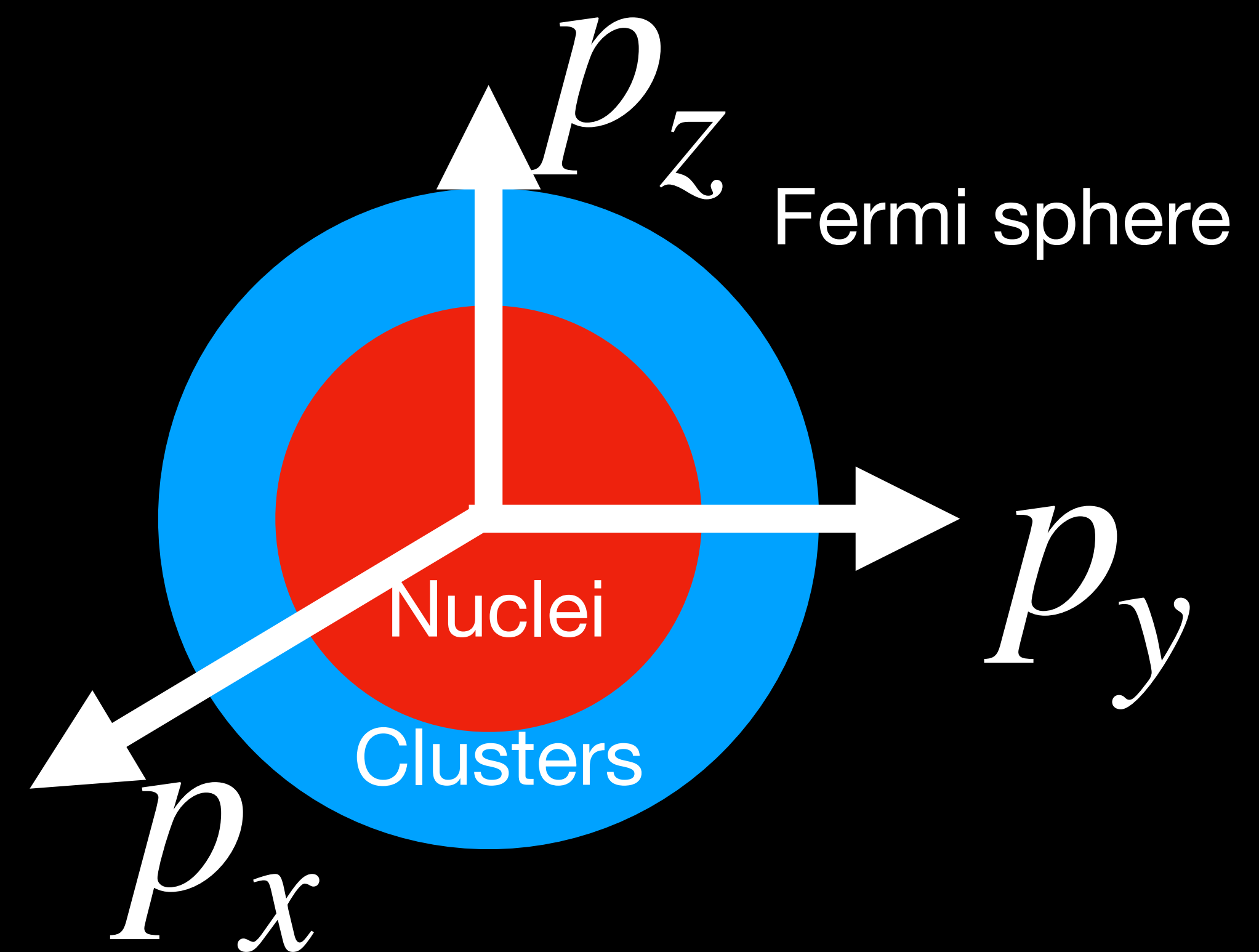
$$Z^{excl}(T, N, V) = Z(T, N, V - v_0 N) H(V - v_0 N)$$

[D.Rishke, M.Gorenstein, et al. Zeitschrift für Physik C Particles and Fields 51 (3)]

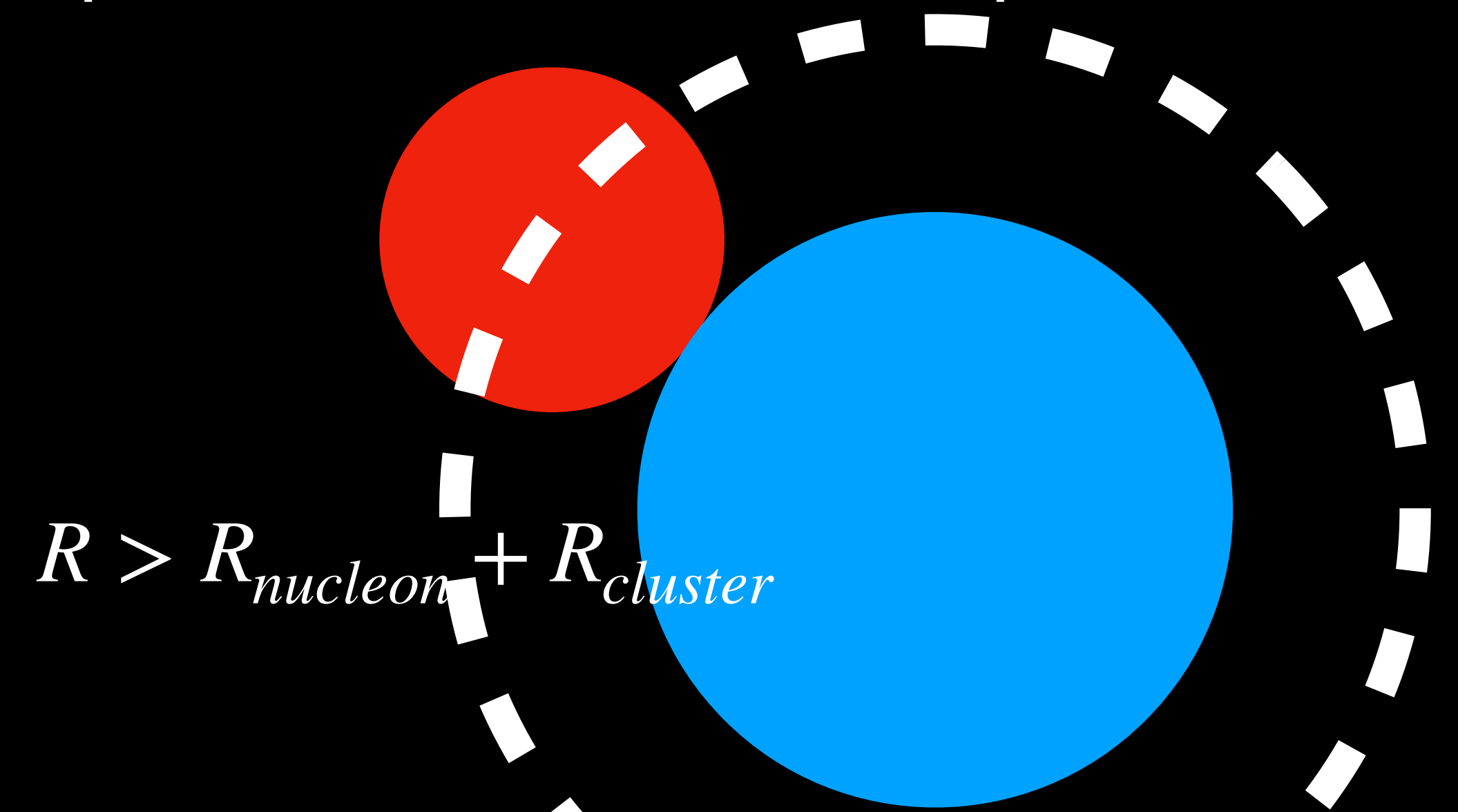
Residual interactions

Chemical non-equilibrium

How presence of resonances affects cluster production



Separated in coordinate space



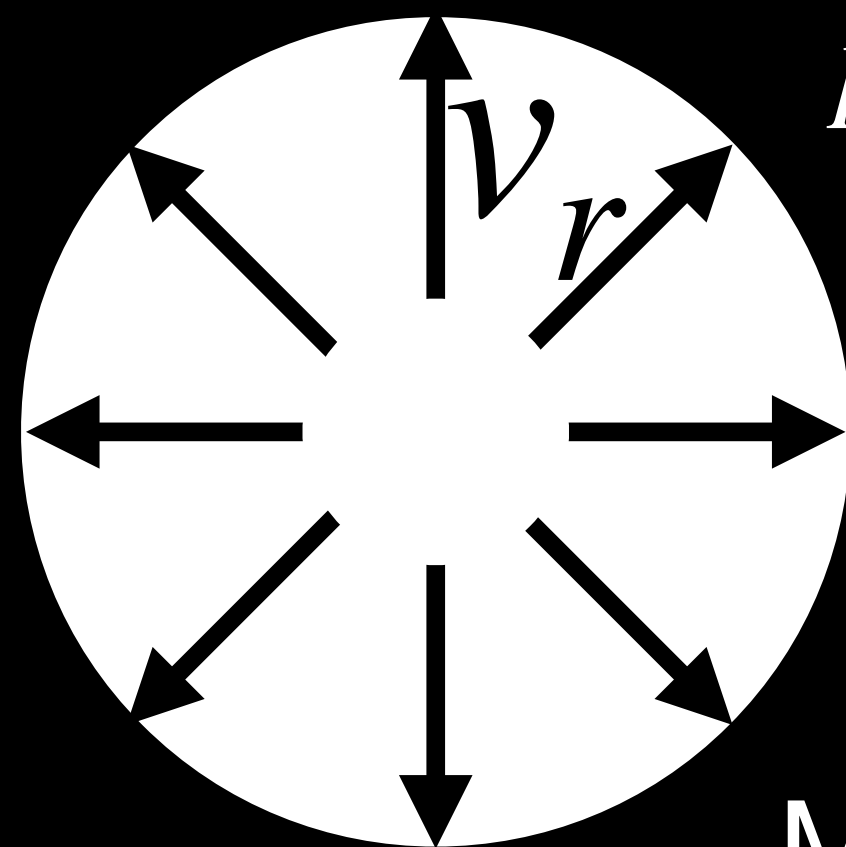
Proton transverse momentum

Clustering changes shape of proton spectra

Tentative signal of temperature increase

Light fragments and free protons add up to the proton spectra before clustering

Spectra includes both contributions from temperature and radial expansion



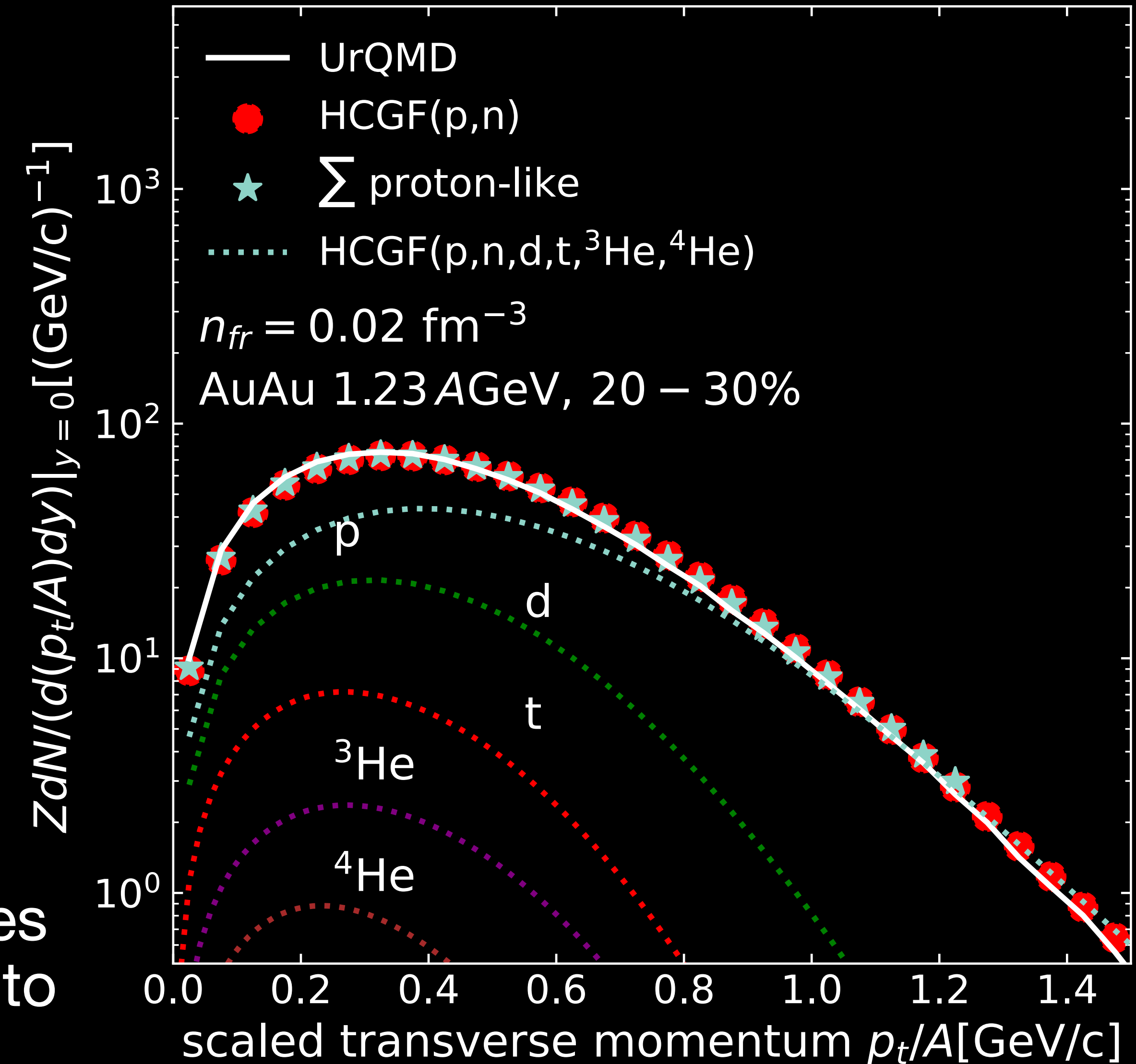
$$p \approx p_{coll} + p_{th}$$

$$p_{coll} \approx mv_r$$

$$p_{th} \sim \sqrt{mT}$$

Heavier particles
more sensitive to
the flow

Mass scaling indicates same source for different particle species



Particle yields

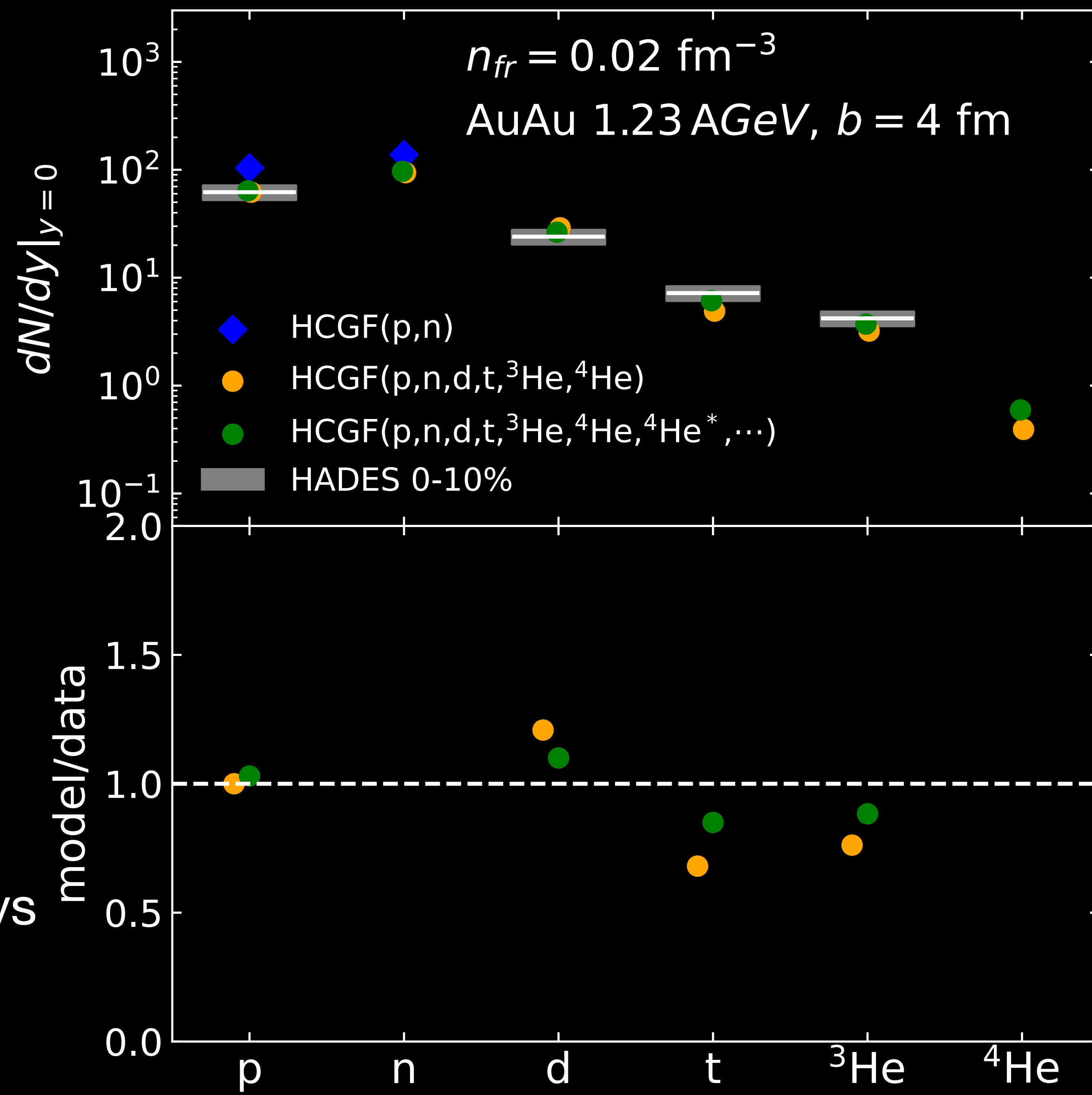
One advantage of our approach is straightforward implementation of additional cluster states.

Including excited states of helium, lithium and beryllium isotopes improves the description of the data.

This effects are expected to amplify at lower energies where majority of protons is clustered.

$\frac{p \times t}{d^2}$ Almost constant without decays

Decays + multiple sources required

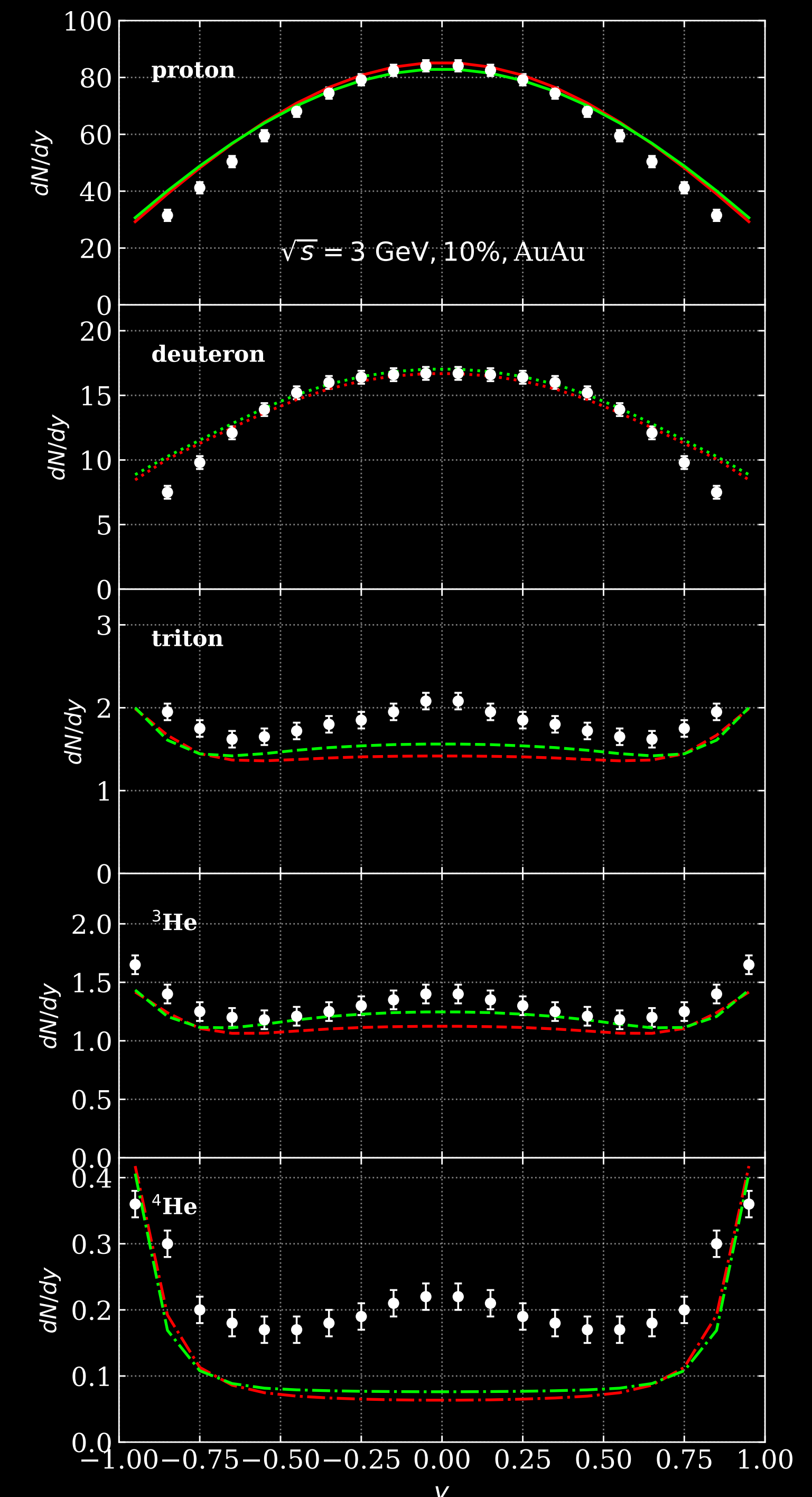
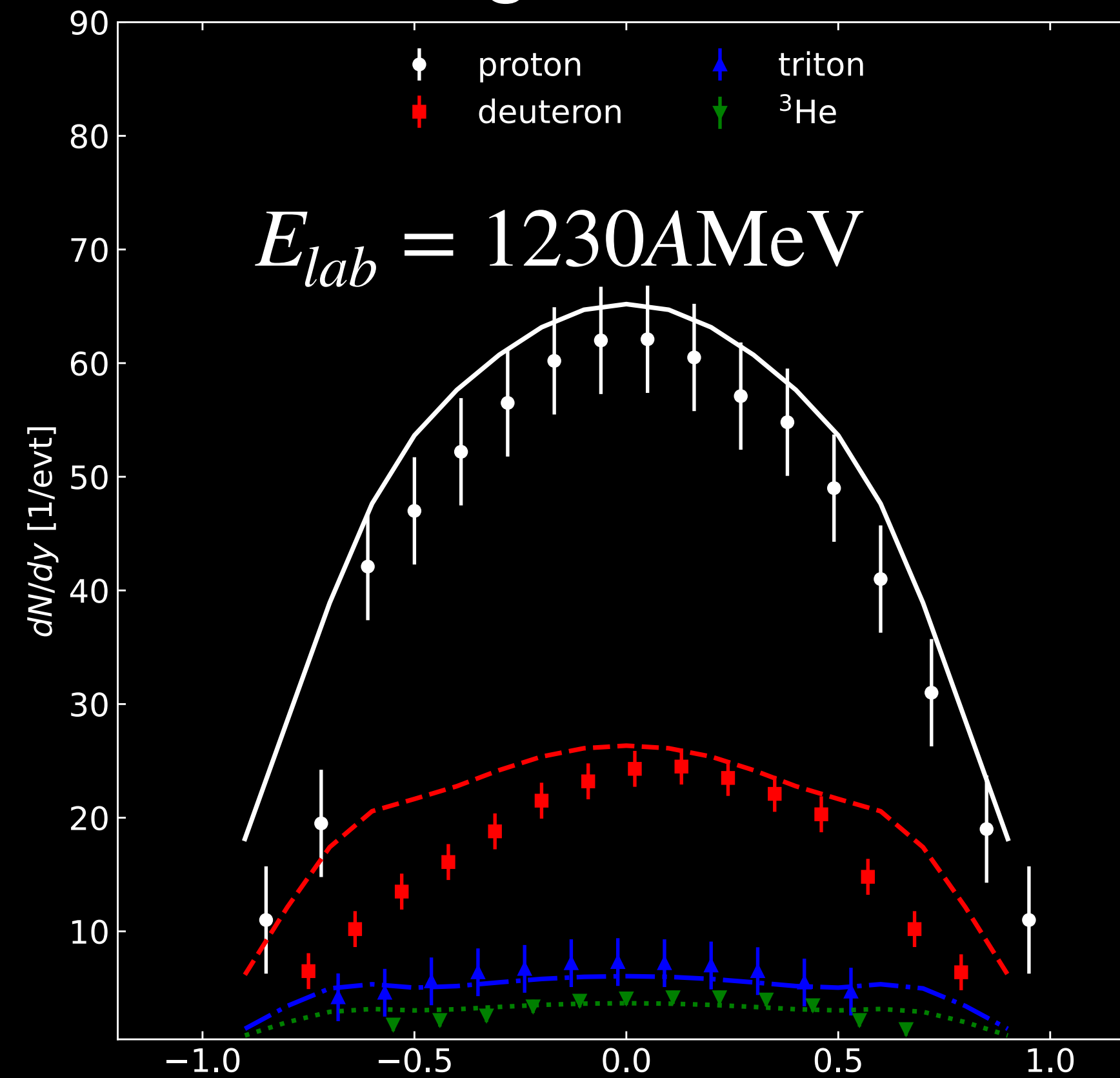


Rapidity spectra

Thermal distribution is best demonstrated with rapidity spectra

Cluster fraction increases as one moves away from mid-rapidity

Transition from central reaction region towards colder spectators



Preliminary $S\pi RIT$ data

Total multiplicity integrates to proton and neutron number before collision

Data has significantly less particles

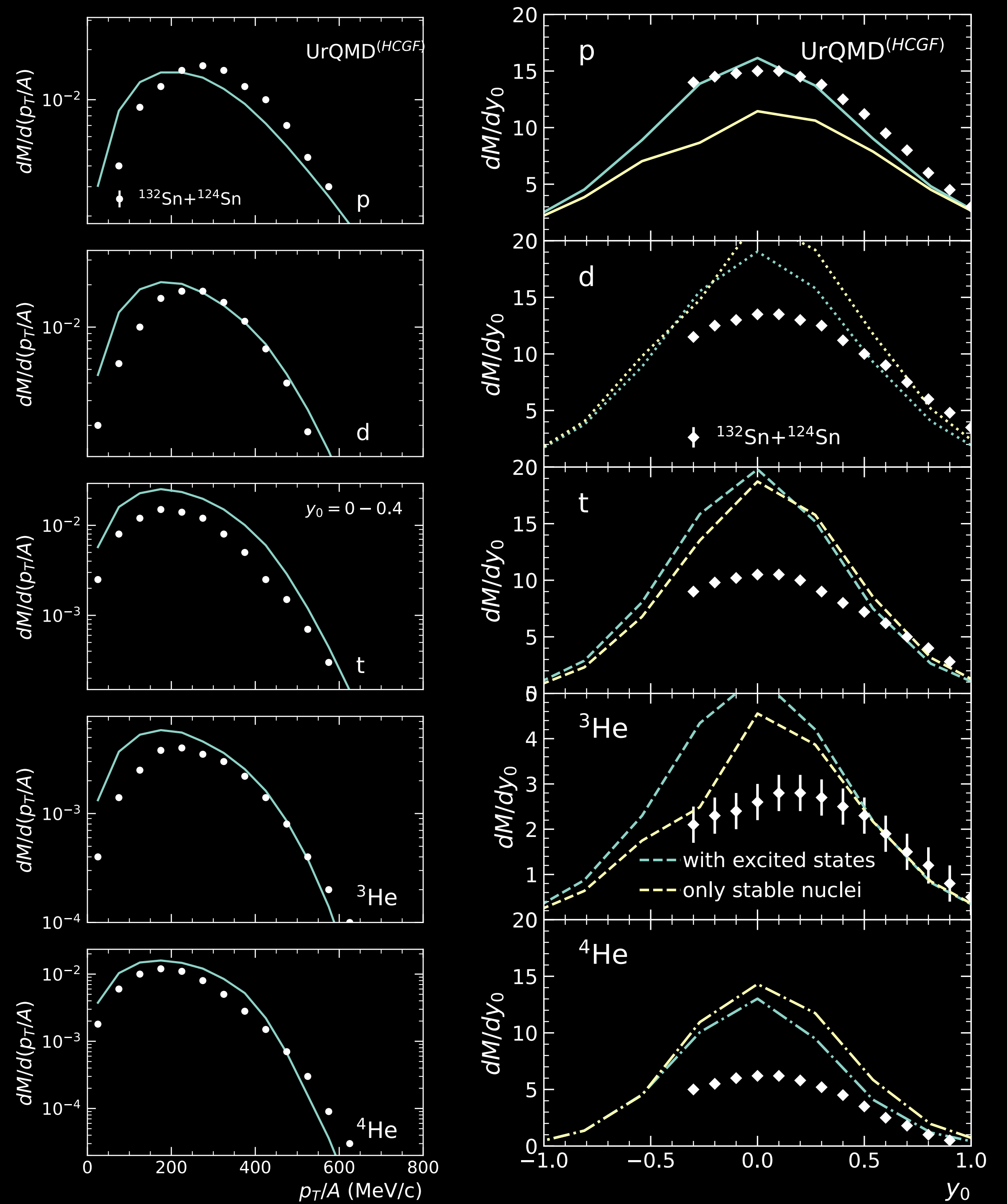
Stopping is too strong in the model?

Heavier stable fragments?

Symmetry energy effects?

Requires isospin treatment and adjusting crosssections?

Impact parameter is wrong?



Azimuthal flow after clustering

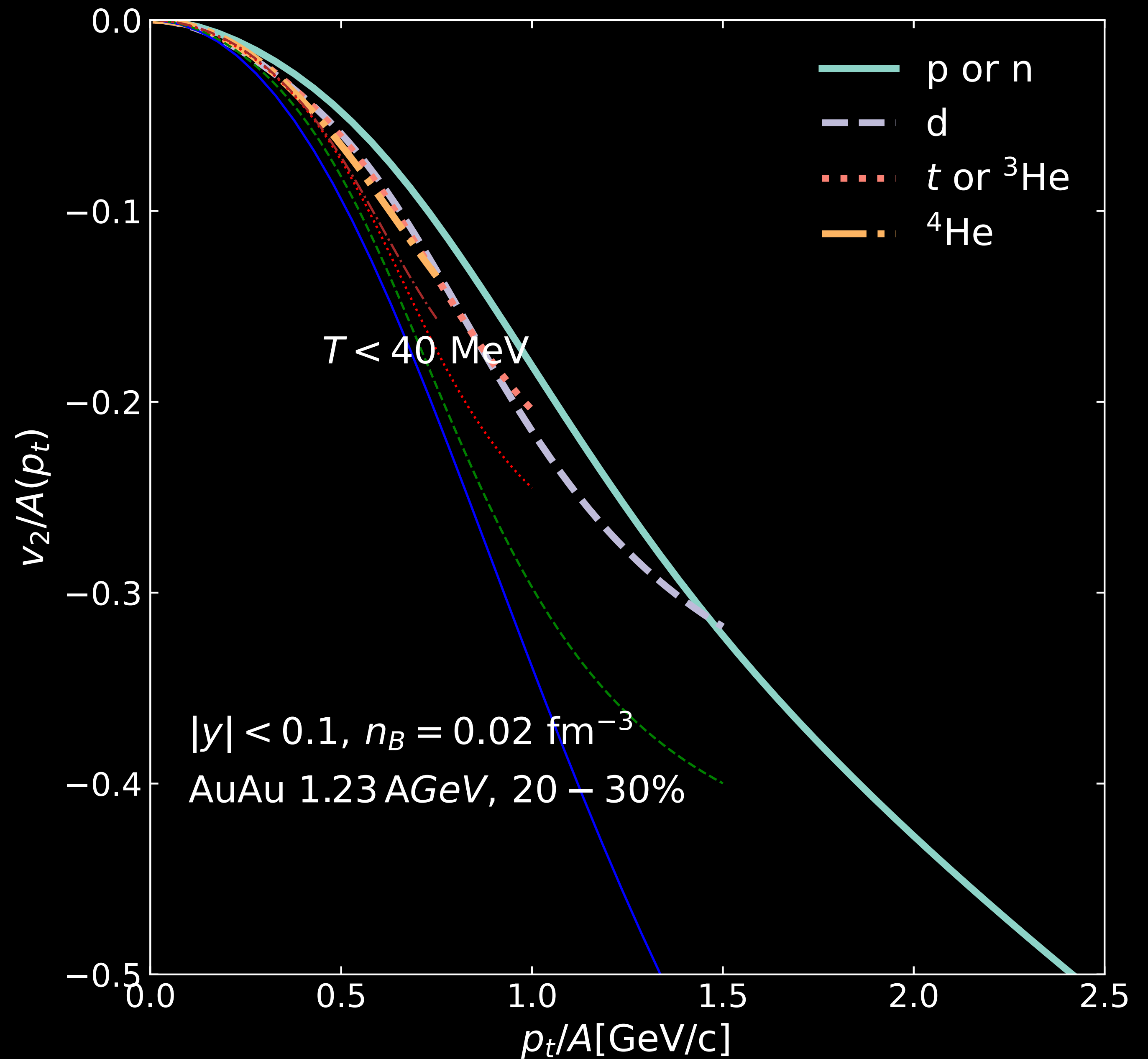
The scaling approximately holds

Time evolution of flow can be seen by considering only emission at lower temperature (later time)

Clusters mostly emitted when freeze-out temperature is low

Clustering weakly changes proton flow

Effects increase at lower energies



Changes in free proton flow

Clustering changes free proton flow

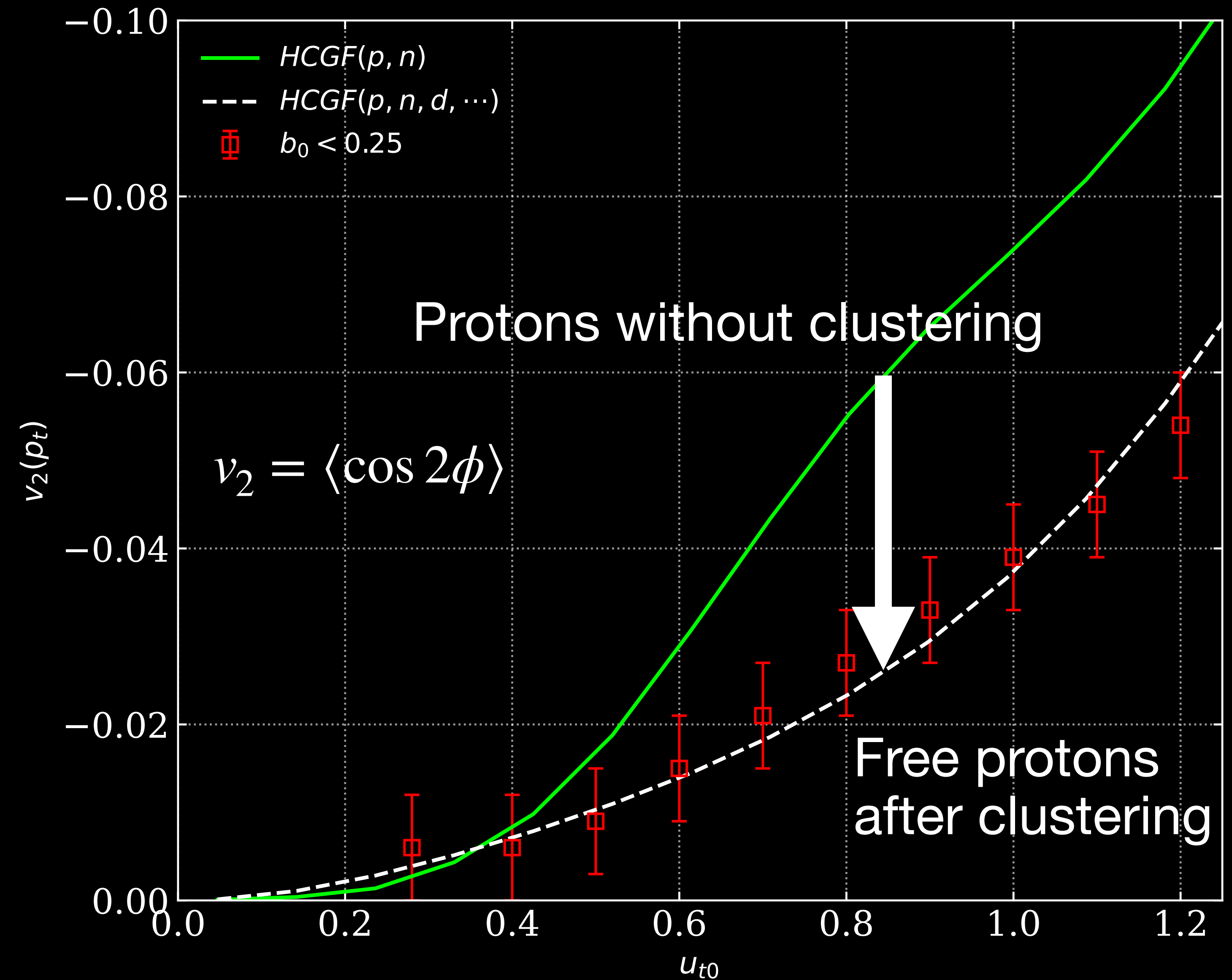
This effect is most present at low momentum where clusters are formed

This modification decreases with energy

This effect is tight to the emission asymmetry between clusters and protons

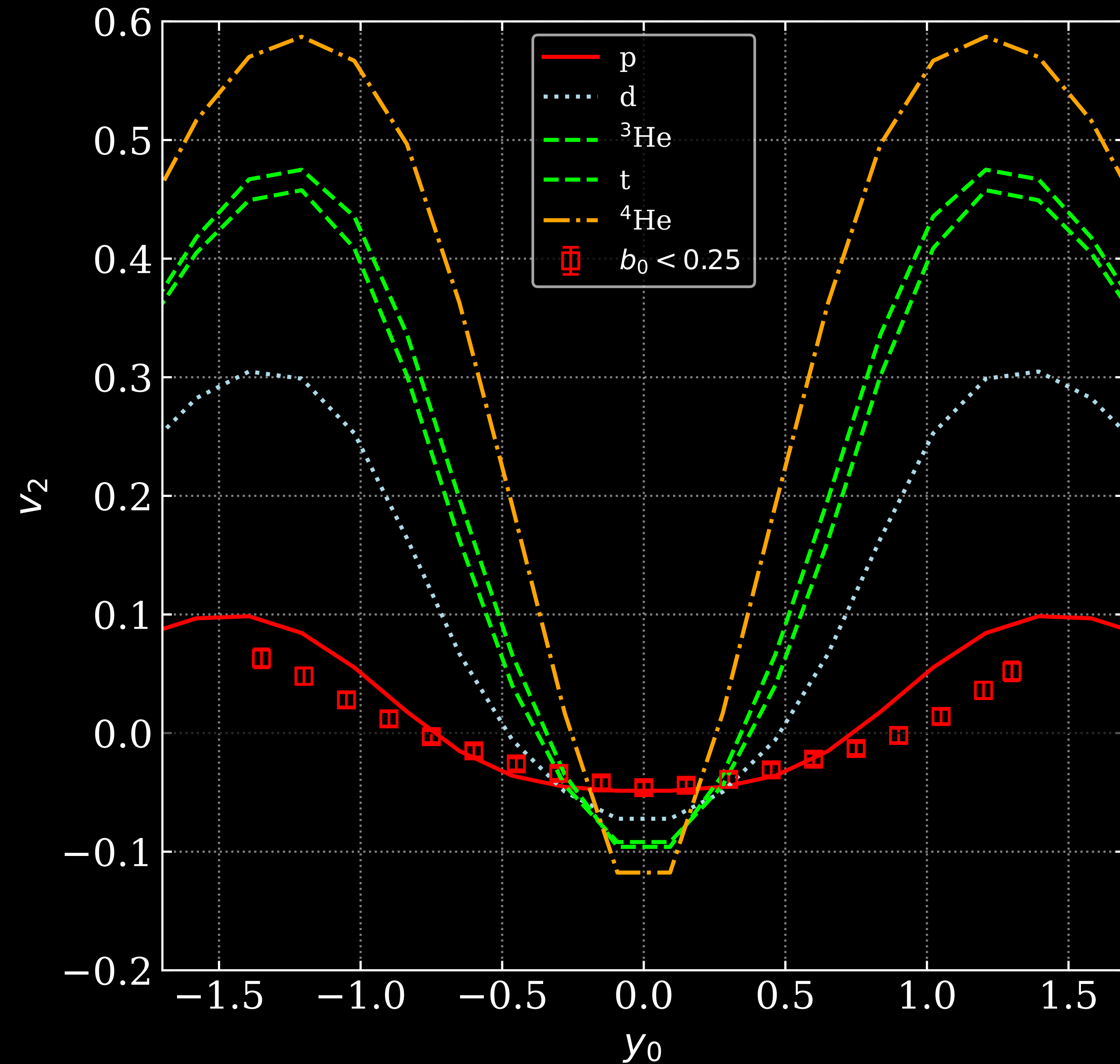
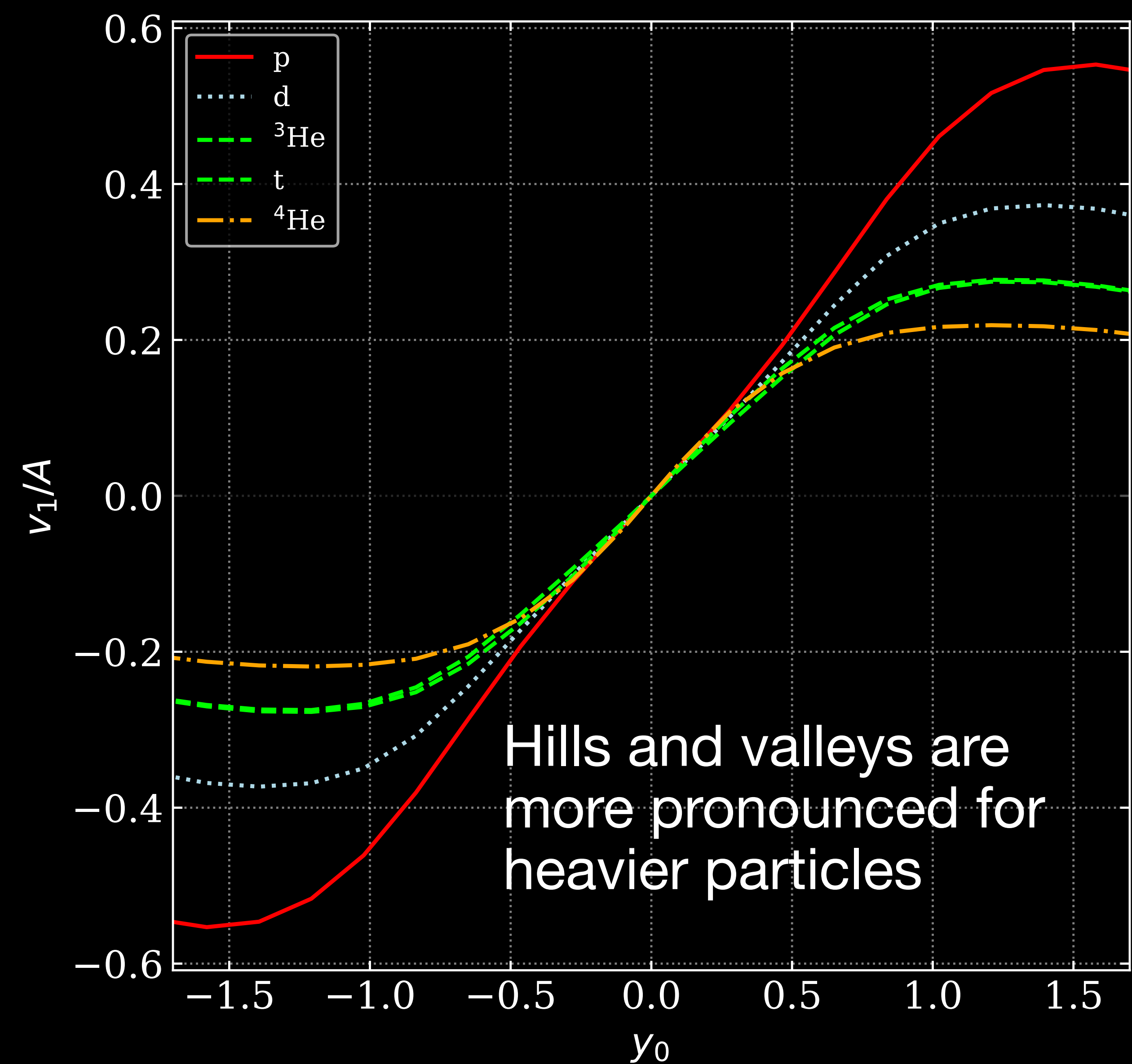
Most protons emitted when freeze-out surface is hot while clusters wait for it to cool down

Proton flow @ 0.6A GeV



Clusters are more sensitive to the flow

29

FOPI Data @ 0.6A GeV ($u_{t0} > 0.4$)FOPI Data @ 0.6A GeV ($u_{t0} > 0.4$)

Conclusions

A novel approach to cluster formation is suggested based on local equilibrium

Spectra and multiplicities of light fragments were calculated in decent agreement with the data

Many observables used in EoS determination are affected by clustering

May improve Bayesian estimates of the equation of state due to clusters being more sensitive to collective flow

